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Saudi Aramco

Engineering Encyclopedia

Saudi Aramco DeskTop Standards

FUNDAMENTALS OF POWER SYSTEM PROTECTION AND COORDINATION

Note: The source of the technical material in this volume is the Professional Engineering Development Program (PEDP) of Engineering Services.

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INFORMATION

The primary objective of a power system protection and coordination study is to select and set equipment protective devices (i.e., fuses, breakers, and relays) to limit the extent and duration of service interruption, whenever equipment failure, human error, or adverse acts of nature occur on any portion of the system. Secondary objectives of the same study are to prevent injury to personnel and to minimize the damage to the power system components.

This Module, which serves as the introductory module to the entire course, has the following objectives:

- To describe the five major components of a protection system (input quantities, circuit breakers, dc station battery, relays, and protected equipment).
- To describe the characteristics of three types of substation circuit arrangements (radial, loop, and selective systems).
- To explain the purposes and characteristics of the one-line diagram, which is the “road map” for the entire electrical power system.
- To explain the general procedures and data requirements that are needed by the engineer to protect and coordinate an electrical power system.

MAJOR COMPONENTS OF A PROTECTION SYSTEM

The purpose of a protection system is to maintain electrical service under normal conditions, and, under abnormal conditions, to minimize damage, to isolate the problem area, and to quickly restore power. Figure 1 shows the five major components (subsystems) of a protection system. Failure of any of the following components usually results in failure of the entire protection system:

- Input quantities (sensors), such as current transformers (CTs) and voltage transformers (VTs).
- Circuit breakers (actuators).
- DC station battery (source of trip power).
- Relays (decision makers, i.e., “brains”)
- Protected equipment such as lines and cables, generators, transformers, motors, and buses.

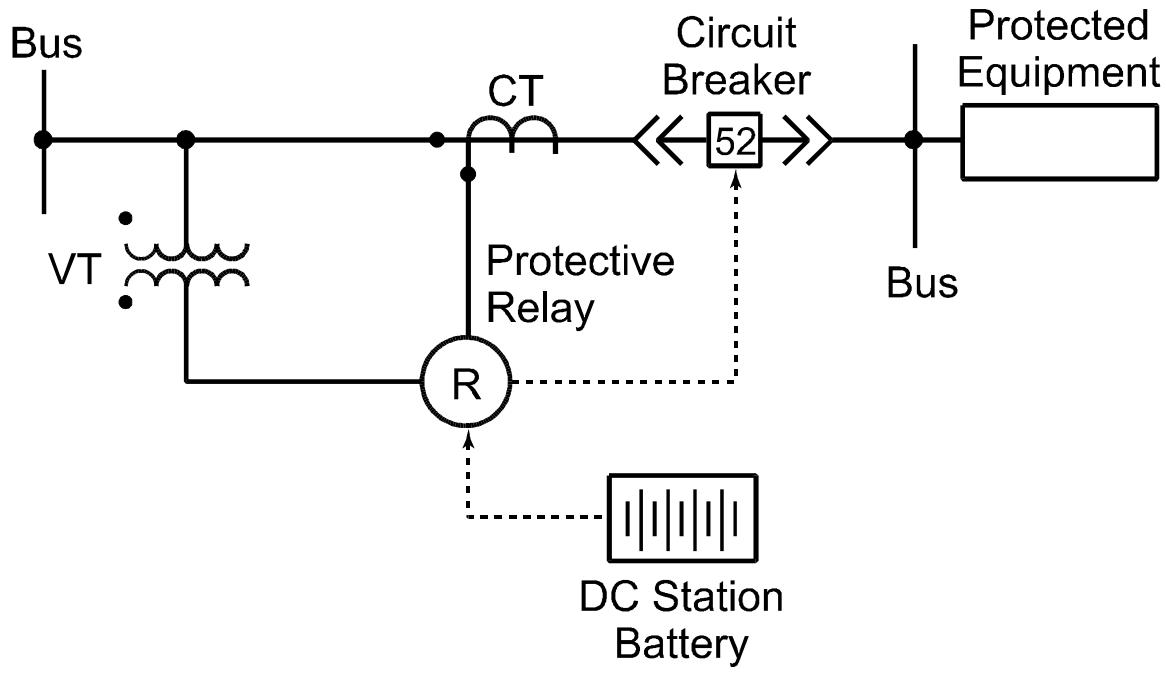


Figure 1. Protection System Components (Subsystems)

Input Quantities

Instrument transformers are used both to protect personnel and apparatus from high voltage, and to allow reasonable insulation levels and current-carrying capacity in relays, meters, and instruments such as voltmeters, ammeters, wattmeter's, and time/overcurrent relays. Instrument transformer performance is critical in protective relaying, because the relays are only as accurate as the instrument transformers.

Instrument transformers may be voltage transformers (VTs), which were also formerly referred to as potential transformers (PTs), or current transformers (CTs). The performance of VTs and CTs are critical to the performance of the above listed devices (e.g., meters and relays). Figure 2 illustrates how instrument transformers are used (connected) in a typical circuit. In the United States, standard instrument transformers are rated at 60 Hz and 5 amperes for the CT secondary output and at 120 volts for the VT secondary output.

Current Transformers (CTs)

The major criterion for selecting a CT is the continuous current rating of the protective equipment and the secondary winding of the CT itself. In general practice, with normal load current flowing through the phase relays, the ratio of a CT is selected so that the secondary current output is 1/2 to 2/3 of 5 amperes at the maximum primary load current. Based on this selection criterion, the CT is usually sized at 150-200% of normal full-load amperes. Where delta-connected CTs are used in the protection scheme, for example, for differential protection of delta-wye connected transformers, the $\sqrt{3}$ factor must be included in the CT ratio selection process. Figure 3 is an illustration of a window-type CT that is used in a modern-day protection system.

Voltage Transformers (VTs)

Voltage transformers (VTs), which in USA were formerly called potential transformers (PTs), are typically selected according to two criteria: the system voltage level and the basic impulse level (BIL) that is required by the system on which the VTs are to be used. The two nominal secondary line-to-line voltages for VTs are 115 and 120 volts; the corresponding line-to-neutral voltages are 66.4 V (115/$\sqrt{3}$) and 69.3 V (120/$\sqrt{3}$). Most protective relays have standard voltage ratings of 120 V or 69 V, depending on whether they are to be connected line-to-line or line-to-neutral. Figure 4 is an illustration of a VT that is used in a protection system.

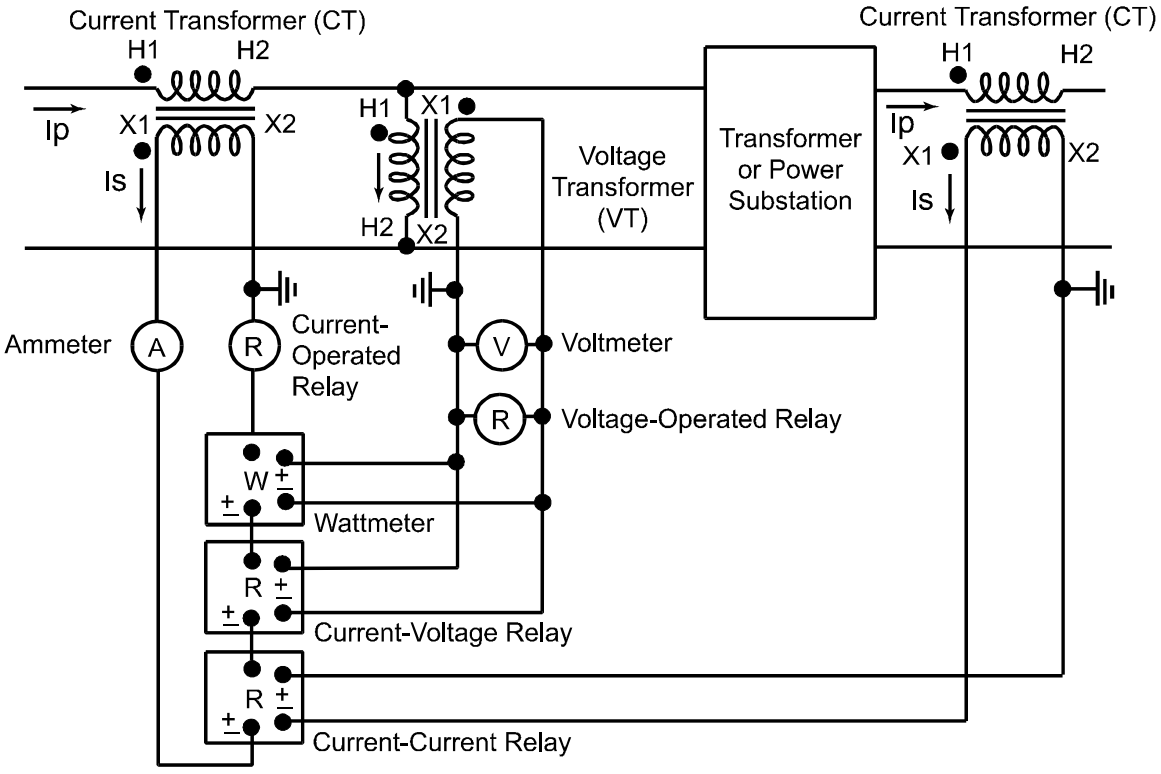


Figure 2. Instrument Transformer Diagram

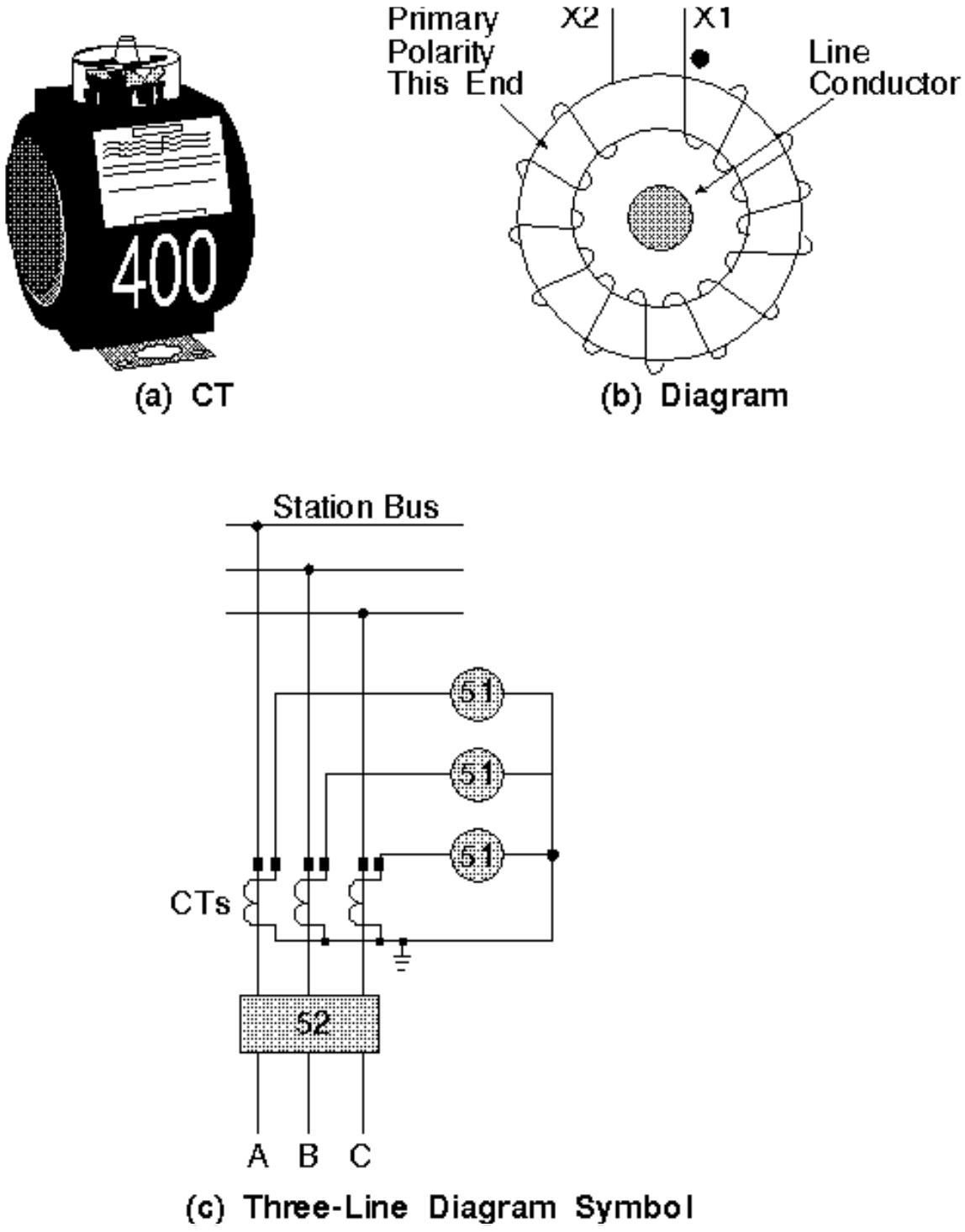
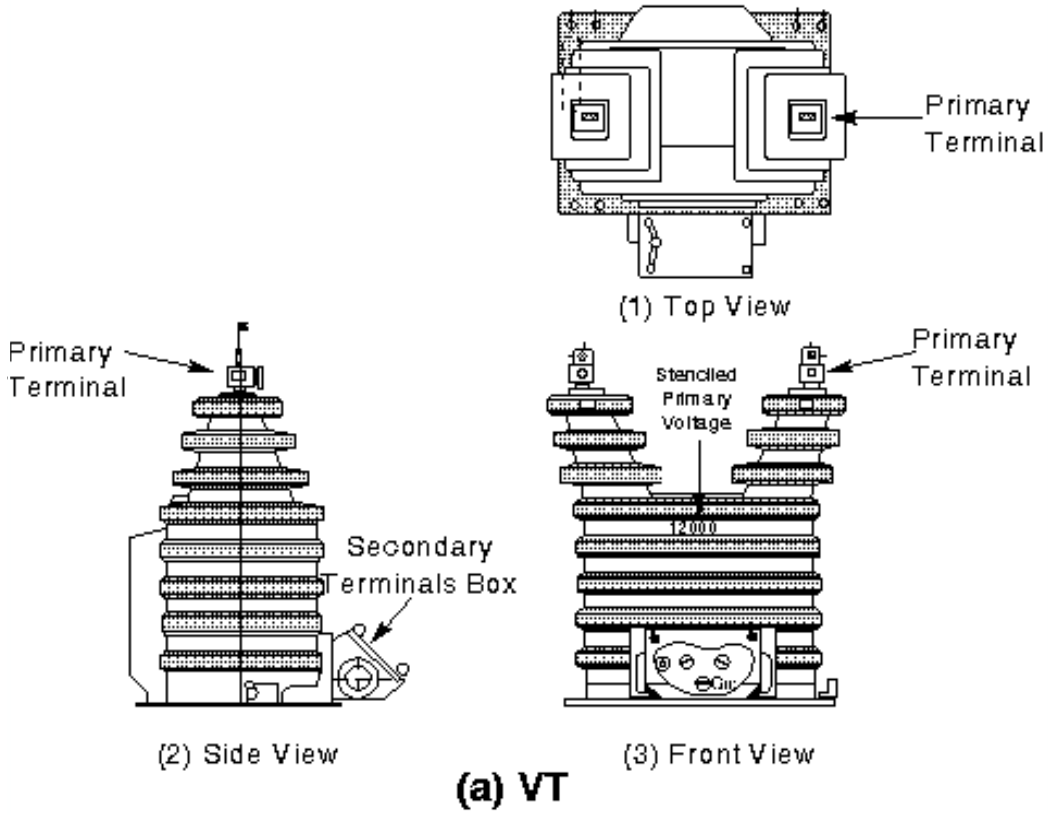
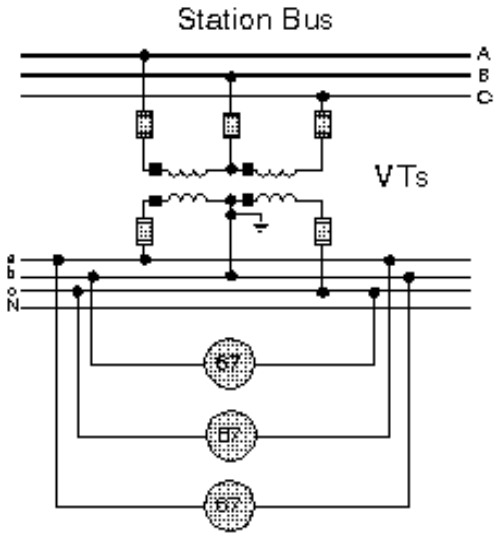


Figure 3. Window-Type Current Transformer (CT)



(a) VT



(b) Three-Line Diagram Symbol

Figure 4. Voltage Transformer (VT)

Circuit Breakers

Although the protective relays are the “brains” of the protection system, they are low energy devices and, therefore, are incapable of clearing or isolating the problem (abnormal condition) that exists in the power system. The circuit breaker (Figure 5) is the device (“the muscle”) that actually isolates the problem (fault). Modern-day medium voltage circuit breakers can interrupt fault currents in the order of 100 kA at system voltages up to 800 kV. In many cases, the breaker clears the fault at the first current zero after the initiation of the fault. However, most medium voltage breakers, in the range of voltages found on Saudi Aramco installations, tend to clear the faults approximately 5 cycles (0.083 sec) plus relay operating time after fault initiation. The circuit breaker is operated by energizing its trip coil from the station battery, and the relays energize the trip coil by closing the contacts between the battery and the breaker’s trip coil (Figure 6).

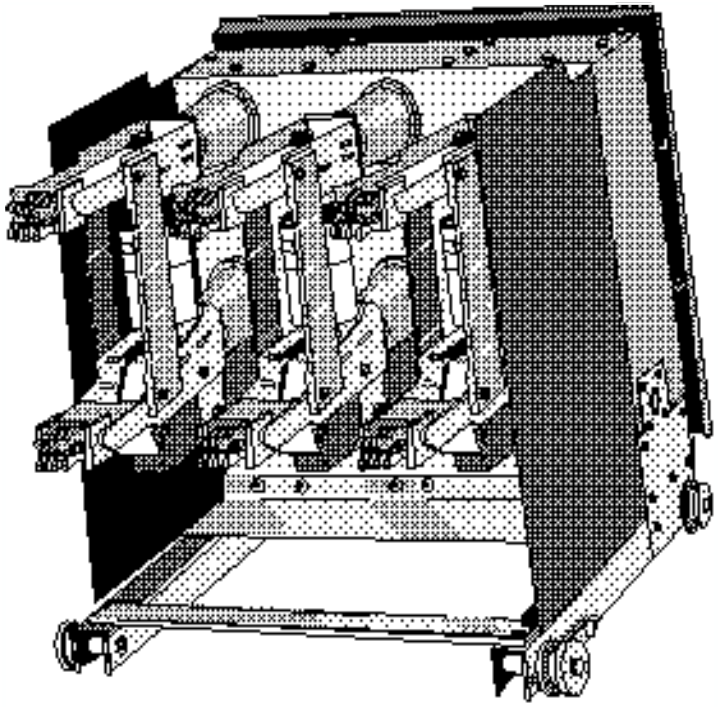


Figure 5. Medium Voltage Vacuum Circuit Breaker (Rear View)

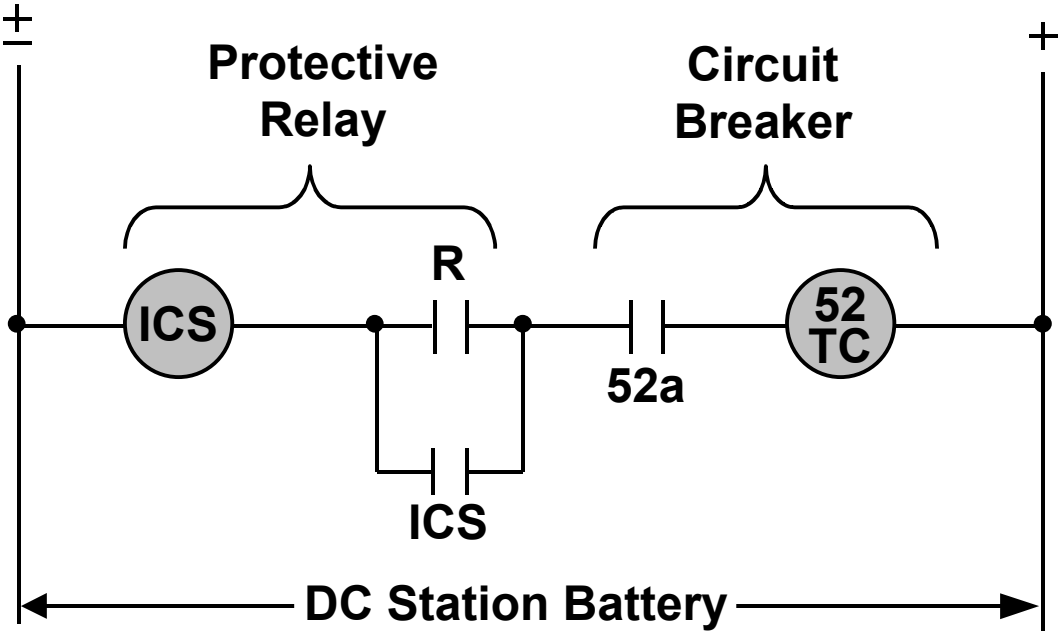


Figure 6. DC Circuit Tripping Schematic

DC Station Battery

Most substations where circuit breakers are installed have a station battery system to supply direct current (dc) to the circuit breaker trip coils (Figure 6), as well as to provide power for emergency alarms, lighting, etc. The dc voltage system in the United States is typically 125 vdc, or 250 vdc for very large substations. With the increasing popularity and use of solid-state electronic relays, 48 vdc also is being used for the protection system. The dc supply system (storage battery) is just as important as any other part of the protection system, and it requires care and maintenance to maintain the protection system's reliability.

Relays

IEEE Standard 100-1984 defines a protective relay as “a relay whose function is to detect defective lines or apparatus or other power system conditions of an abnormal or dangerous nature, and to initiate appropriate control circuit action.” Therefore, protective relays and their associated systems are compact units of analog, discrete component, and/or digital networks that are connected throughout the electrical power system for the purpose of sensing these defective lines or apparatus or other abnormal power system conditions.

Originally, all protective relays were of the electromechanical type, and they are still in wide use today. However, solid-state electronic relays are becoming more common. No one in the electrical power industry doubts that this trend to use solid-state relays will continue, but it will probably be a very long time before the electromechanical relays are superseded or completely replaced in the power system.

Phase Fault Relays

Phase fault relays, along with their ground fault relay counterparts, are the most common type of relays that are used in a protection system. Because fault current magnitudes may range from 5 to 20 times normal full-load amperes, the fault should be cleared as rapidly as possible to minimize the amount of damage to the electrical system and protected equipment. Many fault relays are instantaneous trip devices, which means that the relay trips, without any intentional time delay, upon initiation of the fault. ANSI Devices 50 and 87 are examples of instantaneous type fault relays (Table 1).

Overload Relays

The most common type of relay that is installed in a power system protection system is the time-overcurrent relay, which is an ANSI Device 51 relay. This type of time-overcurrent relay has an intentional time delay built into the analog logic (induction disc) of the relay to permit coordination (selectivity) with other downstream relays. ANSI Device 51 overcurrent relays are also used as phase fault protective relays. Another common type of overload relay is a thermal-overcurrent relay, which is an ANSI Device 49 relay. These thermal- type overcurrent relays are typically available as melting alloy or bimetallic type relays that are used to protect motors under overload conditions.

Ground Fault Relays

Ground fault relays are actually no different than their phase fault counterpart relays. ANSI Devices 51G, 50G, and 87G are used in power systems to protect against ground faults, which are, in many cases, of lower magnitude than phase faults. Ground fault relays are typically set at much more sensitive (lower) settings than the phase fault relays.

Other Relays

ANSI lists approximately 90 different “other” types of relays, ranging from over and undervoltage, reverse power, over and under frequency, and current and voltage unbalance. Table 1 lists several types and functions of the more commonly used power system relays.

Device Number	Definition/Function
27	Undervoltage relay -- a relay that functions on a given value of undervoltage.
46	Phase balance current relay -- a relay that functions on current unbalance, reverse phase-sequence, or negative sequence.
47	Phase balance voltage relay -- a relay that functions on polyphase voltage unbalance or negative sequence.
49	Machine or transformer thermal relay -- a relay that functions when the temperature of a load carrying winding exceeds a predetermined value.
50	Instantaneous overcurrent relay -- a relay that functions instantaneously on an excessive value of current.
51	AC time overcurrent relay -- a relay that functions on an inverse time characteristic.
59	Overvoltage relay -- a relay that functions on a given value of overvoltage.
67	AC directional overcurrent relay -- a relay that functions on a desired value of AC current flowing in a predetermined direction.
74	Alarm relay -- a relay that operates a visual or audible alarm.
86	Locking-out relay -- a relay that functions to shut down and hold equipment out of service.
87	Differential protective relay -- a relay that functions on a percentage or other quantitative difference of two electrical currents.

Table 1. ANSI Standard C37.2-1987 Device Numbers and Functions

Protected Equipment

Lines and Cables

Lines and cables are the backbone of the protected equipment in an electrical power system. If the lines and cables are inadequate, for whatever reasons, unsatisfactory operation of the power system will result no matter how superb the other types of equipment are.

Protection against cable overloads is typically achieved by means of devices that are sensitive to both current magnitude and duration. Protection against cable short circuits is achieved by similar devices, but these short circuit devices are sensitive too much greater current magnitudes and shorter time durations.

Generators

Industrial power systems may include generators as a local source of energy. These generators supply all or part of the total energy required, or, in many cases, they provide emergency power in the event of a failure of the normal source of energy.

Generator protection requires the consideration of many abnormal conditions that are not present with other types of system equipment. Where the generator is unattended, it should be provided with automatic protection against all harmful conditions. In those installations where an attendant is present, it may be preferable to alarm on some abnormal condition rather than remove the generator from service. Generator protective schemes will vary depending on the objectives to be achieved.

Typical generator protection schemes include phase and ground fault protection, reverse power protection, current unbalance protection, and loss of field protection. As the size and importance of the generator increases, more sensitive and complex schemes are used to protect generators.

Transformers

Transformer failure may result in loss of service; however, prompt fault clearing from the system, in addition to minimizing the damage and cost of repairs, usually minimizes system disturbance, the magnitude of the service outage, and the duration of the outage. Prompt fault clearing also will usually prevent catastrophic damage. Proper protection is important for transformers of all sizes, even though transformers are among the simplest and most reliable components in the plant's electrical system.

Small transformers that are rated under 5 MVA are typically protected by use of overcurrent relays (ANSI Devices 50 and 51) for both phase and ground fault protection. As the transformer sizes increase, more elaborate relay schemes are designed and installed to protect the transformer. For example, differential relays are almost always used to provide sensitive fault protection for transformers that are rated 5 MVA and larger. Transformers are also provided with inherent (manufacturer installed) protective devices such as thermal protection (resistance temperature detectors), temperature indicators, and sudden pressure relays.

Motors

There are many variables involved in choosing motor protection: motor importance, motor horsepower rating, and type of motor controller. Therefore, it is recommended that protection for each specific motor installation be chosen to meet the requirements of the specific motor and its use. After the types of protection have been selected, manufacturers' bulletins should be studied to ensure proper application of the specific protection chosen. Typical protection schemes for motors include undervoltage protection, phase and unbalance protection, resistance temperature detector (RTD) thermal protection, locked-rotor protection, and ground and phase fault protection.

Buses

To isolate faults on buses, all power source circuits connected to the bus are opened electrically by relay action or by direct trip device action on circuit breakers, including normally closed bus tiebreakers. Opening of the breaker shuts down all loads and associated processes supplied by the bus and it may affect other parts of the power system as well. When bus protective relaying is used, it should operate for bus or switchgear faults only, because false tripping on external non-bus faults is intolerable. In view of the disastrous effects of a bus fault, the bus equipment should be designed to be as nearly “fault proof” as practicable. High-speed protective relaying should be used to keep the duration of the fault to a minimum. These factors limit the damage and minimize the effects on other parts of the power system. When medium voltage industrial power systems are grounded through resistance to limit ground fault damage, as they are on Saudi Aramco installations, the current available to detect a ground fault is small, and, therefore, the protective relaying should be very sensitive. Bus overcurrent protection (overload and phase fault) is typically provided by time/overcurrent relays (ANSI Device 51). Where more sensitive and high speed fault protection is desired, differential protection (ANSI Device 87B) is used to protect the bus and its associated switchgear.

CHARACTERISTICS OF BASIC TYPES OF SUBSTATION CIRCUIT ARRANGEMENTS

In an industrial plant, a variety of basic circuit arrangements are available for the distribution of electrical power. Selection of the "best" system depends upon the need of the process (e.g., crude oil refining) in which the system is to be used. Reliability of the power supply depends upon the load. For example, a simple radial system is probably adequate to supply a housing area, whereas a more sophisticated, expensive, and reliable loop system is more than likely required to supply a critical refinery process. This Information Sheet will briefly describe the following types of substation circuit arrangements: radial systems, loop systems, and selective systems.

Radial Systems

Figure 7a illustrates a simple radial system, and Figures 8b and 8c illustrate expanded radial systems.

Simple Radial System

A simple radial system (Figure 7a) looks like an inverted tree. A single primary service and transformer serve the entire load. There is no duplication of electrical equipment (cables, breakers, etc.), and the system investment is the least expensive of all of the types of circuit arrangements. The operation and expansion of the radial system is simple, and the reliability is high if top-quality components are used. The radial system also is a relatively easy system on which to perform a short circuit or coordination study. Unfortunately, loss of a single cable or transformer, a tripped breaker, or blown fuse will shut down the entire radial system. The equipment must also be shut down to perform routine maintenance. The simple radial system is an adequate power system circuit arrangement for most non-critical process loads.

Expanded Radial System

The expanded radial system (Figures 8b and 8c) is just an expansion of the simple radial system, and it is used to supply power to multiple unit substations that are near major load centers. The advantages and disadvantages described for simple radial systems also apply to expanded radial systems.

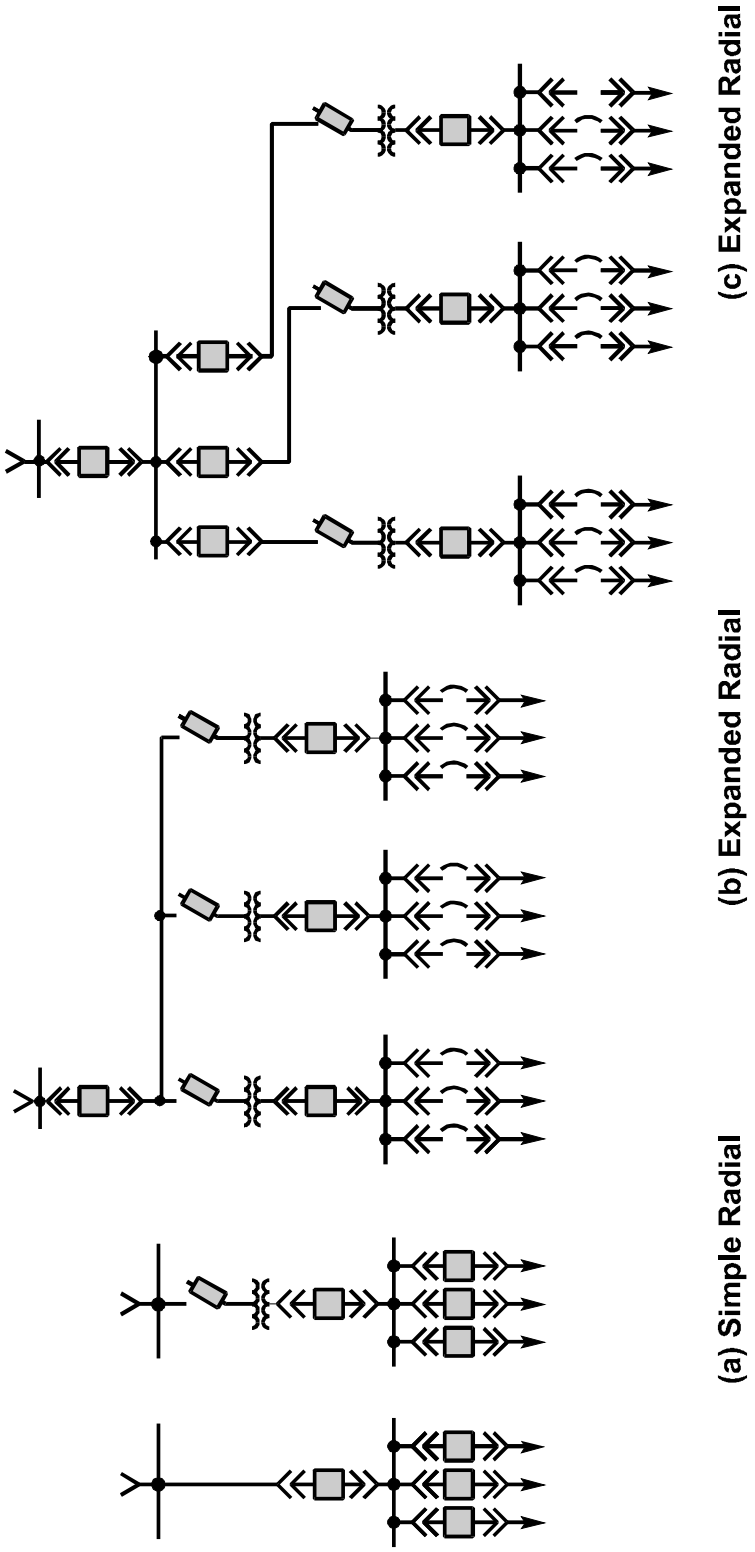


Figure 7. Radial Systems

Interconnected Systems

Figure 8a illustrates an interconnected (loop) circuit arrangement.

An interconnected system is more reliable than a radial system. If one source feeder fails, the other source feeder supplies the load. The system is more dangerous to work on than the radial system because power is supplied from both directions to the load. Fault duties are typically double that of a radial system because the parallel feeders' impedance is 50 percent of the single radial feeder's impedance. Interconnected systems provide greater reliability for critical loads because single faults will not isolate the system. They are also more expensive because of the duplication of equipment. Short circuit studies performed on loop systems are more tedious, and coordination studies more complex because of the need to use directional relays (ANSI Device 67). If any of the normally-closed (N.C.) switches illustrated in Figure 8a are opened, the system reverts to a simple radial system.

Loop System

The system in 9b is described in Saudi Aramco distribution systems as a Loop. It is really a radial system, but provides improved service because the open point(s) can be changed to isolate a faulty cable and maintain supply to the transformer.

Selective Systems

Selective system circuit arrangements are either primary selective, secondary selective, or a combination of both.

Primary Selective System

Loss of a primary source can be protected against by use of a primary selective system (Figure 9), where each transformer is supplied by two sources. Normal operation is to supply half the load (transformers) from one source, with the other source acting as the alternate (emergency) source. Switching of the load (transformers) over to the alternate source can be manual or automatic, but there will be a power interruption until the load is transferred to the alternate source.

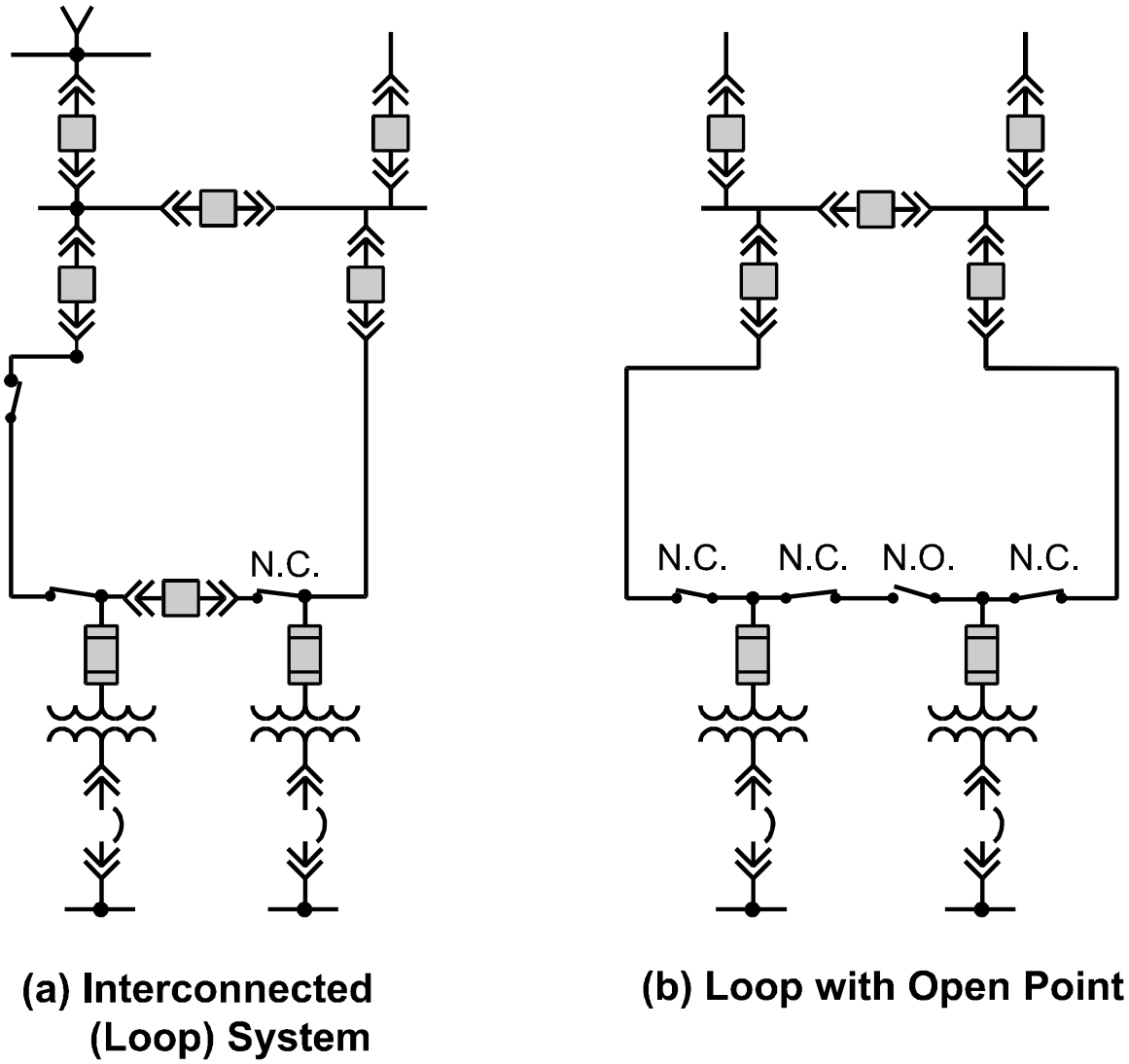


Figure 8. Loop Systems

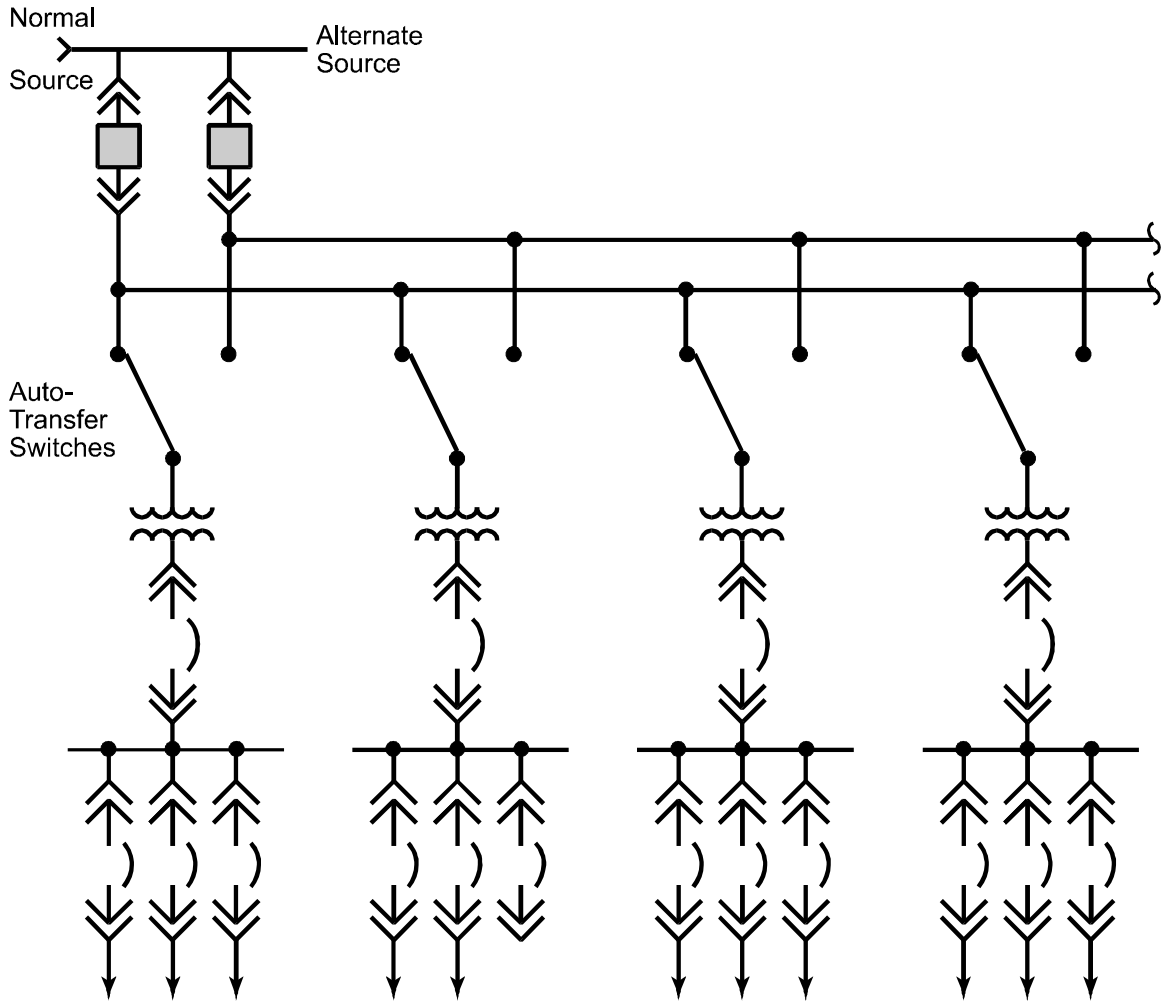


Figure 9. Primary Selective System

Secondary Selective System

If pairs of unit substations transformers are connected through a normally open (N.O.) tie breaker, the system is called a secondary selective system (Figure 10a). Each unit substation normally carries half the load, and in the event of a failure of the normal source, or for routine normal maintenance, the N. O. tiebreaker is closed. The secondary selective system is the preferred Saudi Aramco circuit arrangement for large unit substations. For routine maintenance purposes, Saudi Aramco requires closing of the N. O. tie breaker before opening one of the main breakers to prevent any interruption of power to the load. If the substations are geographically remote from one another, two tiebreakers (one N.O. and one N.C.) are used for selective switching purposes, as illustrated in Figure 10b. If the tiebreakers are normally closed, additional protection (e.g., directional relays) are required to inhibit (prevent) the alternate source from “back feeding” into a fault.

Combined Selective System

The combined selective circuit arrangement system (Figure 11) is simply a combination of both the primary and secondary selective systems. Figure 12 illustrates an alternative type of combined selective system.

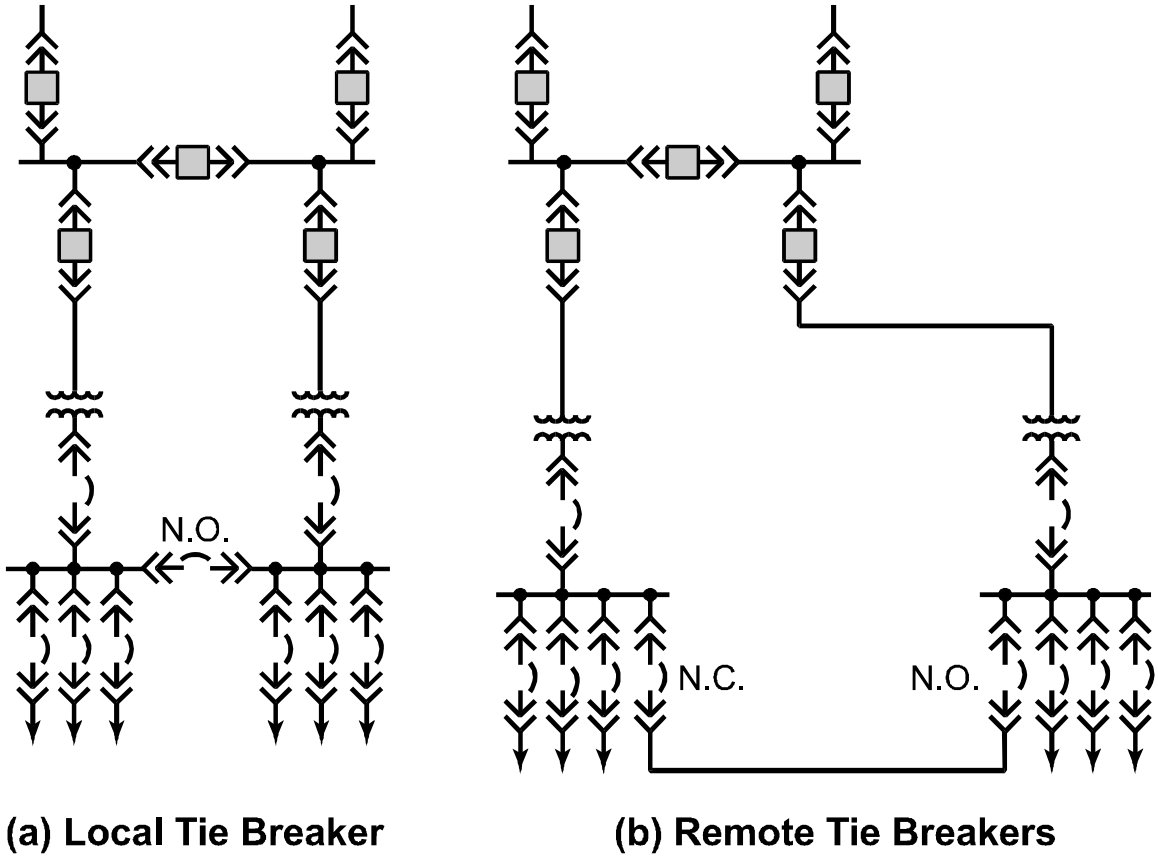


Figure 10. Secondary Selective Systems

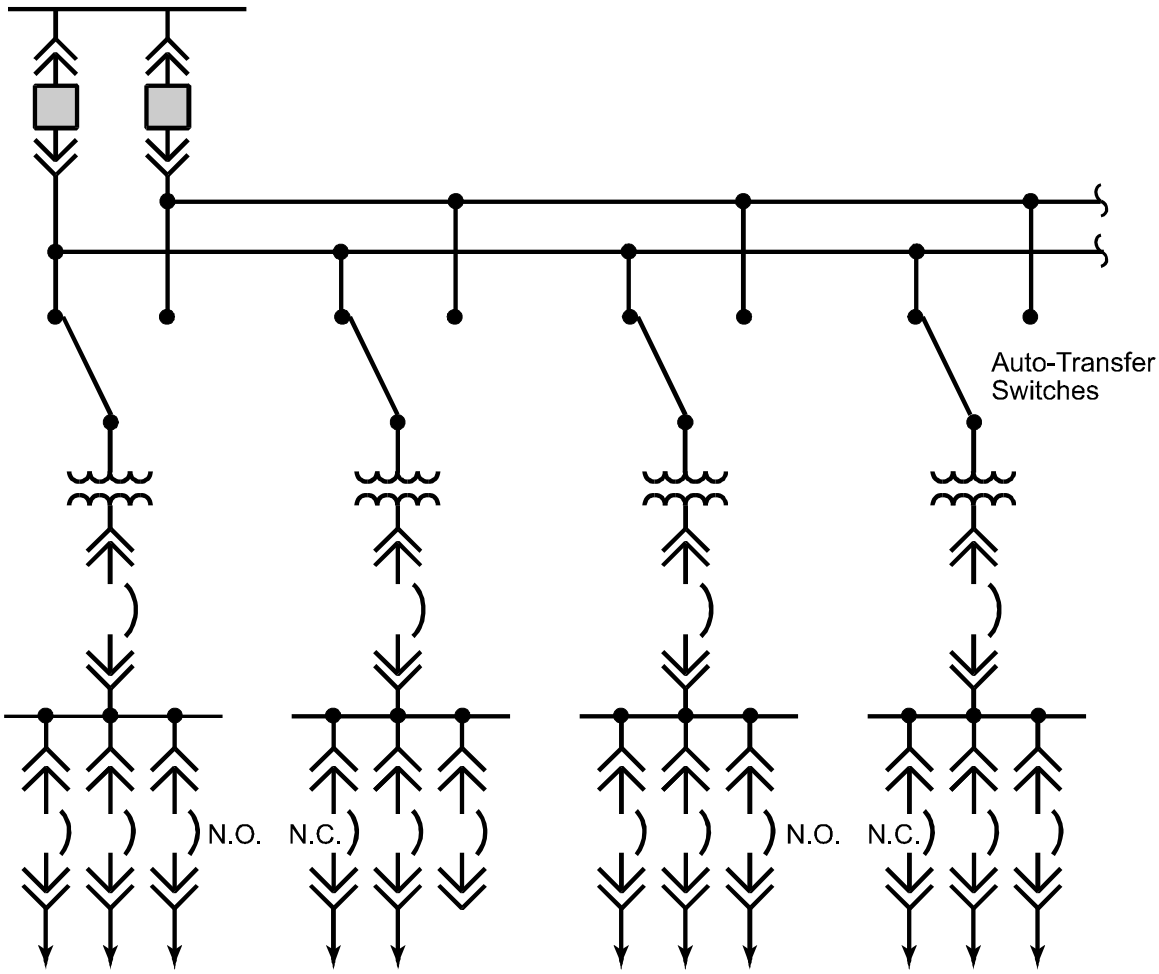


Figure 11. Combined Selective System

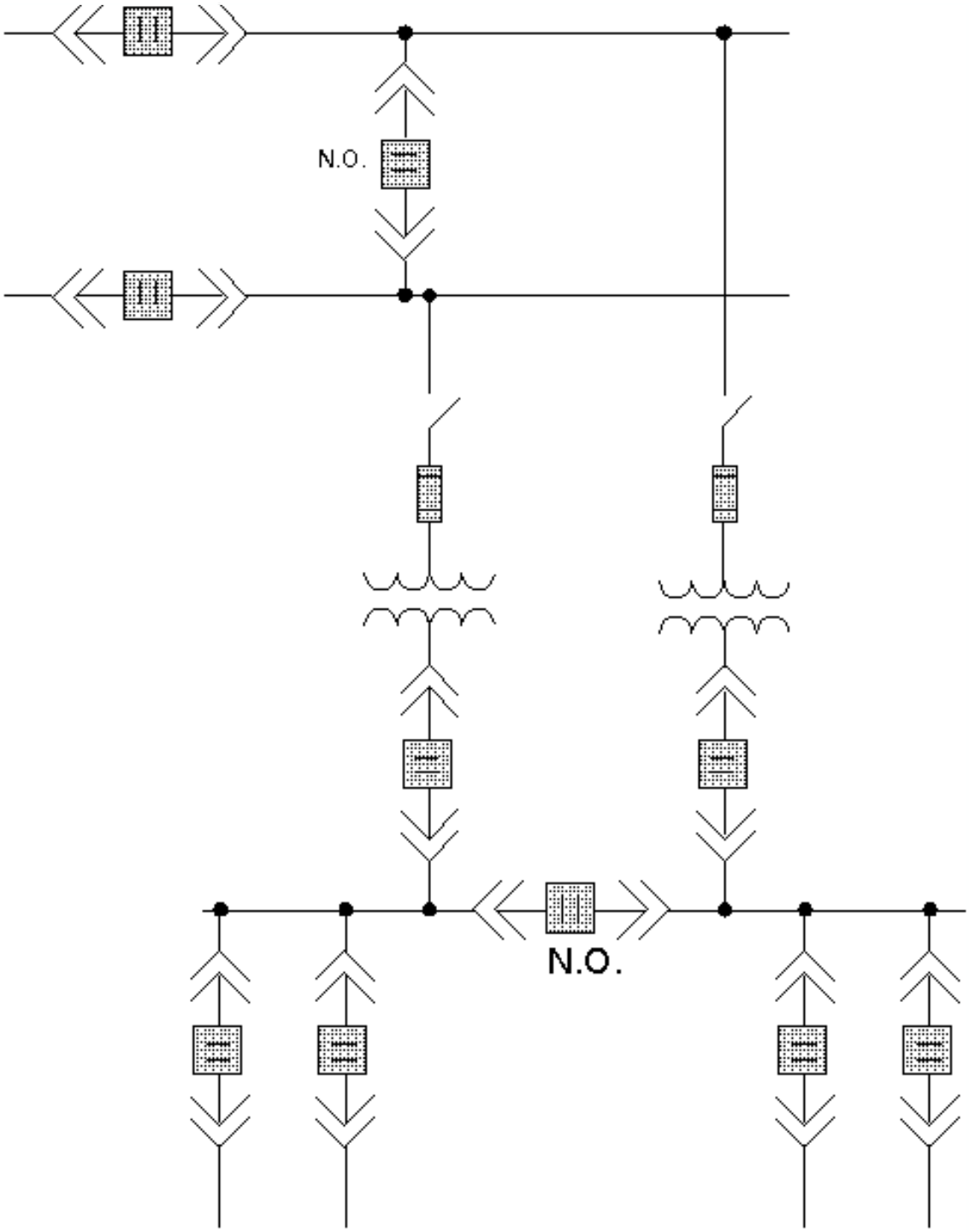


Figure 12 Alternate Combined Selective System

ONE-LINE DIAGRAM: PURPOSES AND CHARACTERISTICS

The most commonly used diagram in an industrial power system is the one-line diagram (Figure 13). This diagram is very useful in showing, by means of standard graphical symbols and nomenclature, an overall power system arrangement. For maximum usefulness, the relative physical arrangement of the electrical system should be shown on the one-line diagram.

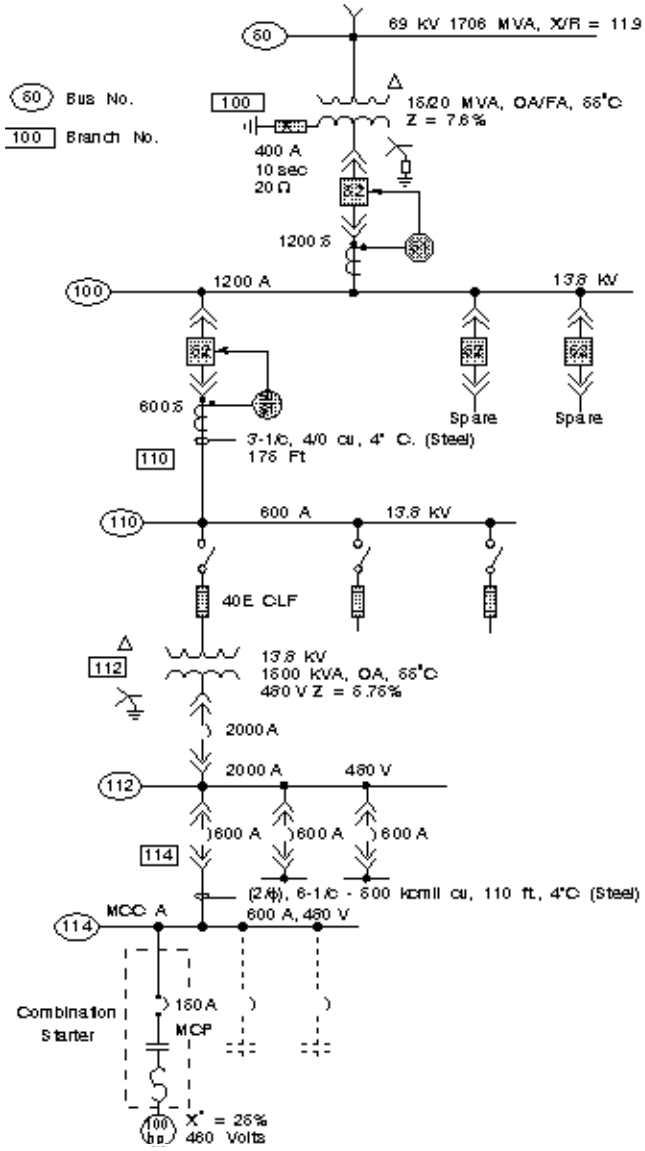


Figure 13. One-Line Diagram

Purposes

Power System Studies

The one-line diagram is most commonly used to perform power systems studies. The following information is typically provided, as a minimum, on the one-line diagram, regardless of the type of power system study that is being performed.

- Bus current and voltage ratings.
- Short circuit current available.
- Voltage and current ratios of instrument transformers.
- Protective device (e. g., circuit breakers and fuses) ratings.
- Functions of relays indicated by ANSI device numbers.
- Ratings, type, and impedance of motors and transformers.
- Connections (e.g., delta or wye) of transformers.
- Number, length, size, and type of conductors and conduit.

The final application of the drawing (e. g., short circuit study, coordination study, and construction) will determine the exact information that exists on the one-line diagram. For example, impedance of a motor is required for a short circuit study, but not for a coordination study. Relay and adjustable settings of circuit breakers are required for a coordination study, but are not required for a short circuit study.

Operations and Maintenance

The one-line diagram is also commonly used by technicians to operate and maintain the plant electrical distribution system. For example, the one-line diagram is used to determine which breakers or switches should be closed or opened to switch to alternate sources of power because of a fault on the system. As another example, the one-line diagram is used to perform “locking and tagging” procedures when equipment is to be removed from service. Both of these uses of the one-line diagram point out that the diagram must be kept up-to-date and accurate. Use of inaccurate data on a one-line diagram that is used to perform a power systems study could result in additional costs, but use of inaccurate data that are used for switching

purposes or locking and tagging procedures could result in loss of life.

Construction

Probably the least common use of the one-line diagram is for construction purposes. Electricians usually will require more details to construct or install electrical equipment than is available on the one-line diagram.

Characteristics

Commonly Used Symbols

The commonly used graphical symbols, when used consistently and in conformance with general practice, provide a valuable tool to the power systems engineer.

Saudi Aramco Drawing No. 990-P-AB036766 describes the standard electrical symbols used for power system one-line diagrams for Saudi Aramco installations. As with most Saudi Aramco standards, the symbols on the drawing are in accordance with the more nationally recognized ANSI/IEEE (American National Standards Institute/Institute of Electrical and Electronics Engineers) Standard 315-1975 (old ANSI Y32.2 1970). Figure 14 lists several of the more common symbols that will be used in this course. ANSI/IEEE Standard 315-1975 was reaffirmed in 1988.

ANSI/IEEE Device Numbers and Functions

Each device identified on the one-line diagram should be given a number in accordance with ANSI Standard C37.2-1987. Some of the more common device numbers and their functions are shown in Tables 2 and 3.

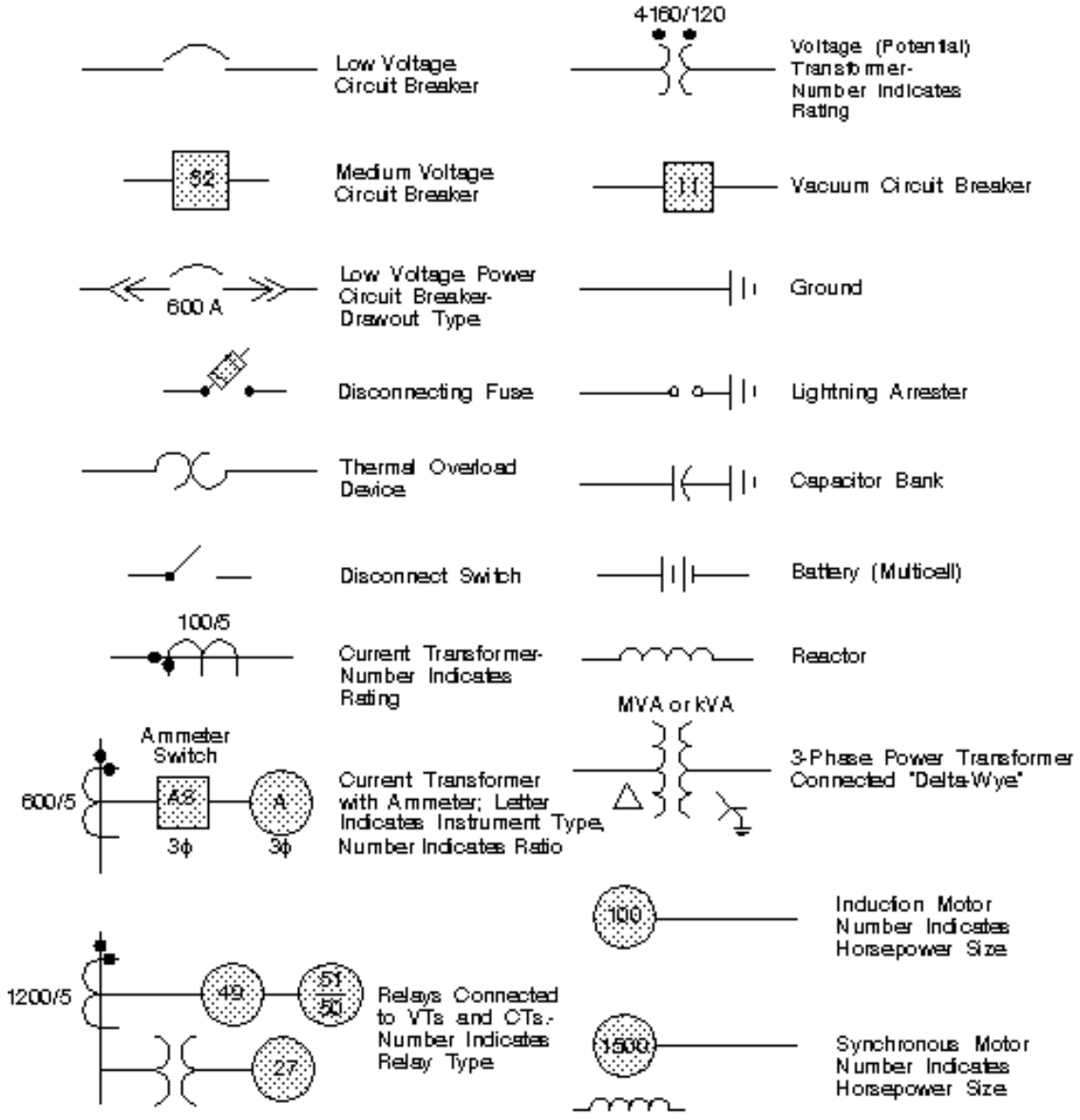


Figure 14. Commonly Used Symbols for One-Line Diagrams

DEVICE NUMBER	DEFINITION/FUNCTION
25	Synchronizing or synchronism – check device – a device that when two ac circuits are within the desired limits of frequency, phase angle, or voltage, to permit or to cause the paralleling of these two circuits.
27	Undervoltage relay – a relay that functions on a given value of undervoltage
46	Phase balance current relay – a relay that functions on current unbalance, reverse phase-sequence, or negative sequence.
47	Phase sequence voltage relay – a relay that functions on polyphase voltage unbalance or negative sequence.
49	Machine or transformer thermal relay – a relay that functions when the temperature of a load carrying winding exceeds a predetermined value. Most commonly used for motor overload function. Thermal overload relays use the heating effect of the load current but do not actually measure the motor temperature.
50	Instantaneous overcurrent relay – a relay that functions instantaneously on an excessive value of current.
51	An ac time overcurrent relay – a relay that functions on an inverse time characteristic on a given value of overcurrent.
52	An ac circuit breaker – a device that is used to close or interupt an ac power circuit under normal or fault conditions

Table 2. ANSI Standard C37.2-1987 Device Numbers and Functions - Part I

DEVICE NUMBER	DEFINITION/FUNCTION
59	Overvoltage relay – a relay that functions on an inverse time characteristic on a given value of overvoltage.
62	Time delay relay – a relay that serves in conjunction with another relay to shutdown, stop, or open an automatic sequence.
63	Pressure switch (sudden pressure relay) – a switch that operates on given values or on a given rate of change of pressure.
67	An ac directional overcurrent relay – a relay that functions on a desired value of ac current flowing in a predetermined direction.
74	Alarm relay – a relay that operates a visual or audible alarm.
81	Frequency relay – a relay that functions on a predetermined value of frequency.
86	Locking-out relay – a relay that functions to shut down and hold equipment out of service
87	Differential protective relay – a relay that functions on a percentage or other quantitative difference of two electrical currents.

Table 3. ANSI Standard C37.2-1987 Device Numbers and Functions - Part II

GENERAL PROCEDURES AND DATA REQUIREMENTS FOR A COORDINATION STUDY

Introduction

An overcurrent protective device time/current (T/C) coordination study is an organized engineering effort to determine the appropriate ampere ratings, types, and settings of the overcurrent protective devices (fuses, breakers, and relays) that are installed in an electrical power system. The objective of the coordination study is to ensure among the devices a T/C coordination that achieves the desired system protection and electrical service continuity goals.

Maximum protection requires that the overcurrent protective devices be rated, selected, and adjusted to allow the normal load currents to flow, while instantaneously opening the circuit when abnormal currents flow.

On the other hand, maximum service continuity requires that the devices be rated, selected, and adjusted so that only the fault current-carrying device nearest the fault opens to isolate the faulted circuit from the system, while permitting the rest of the system to remain in operation. Maximum service continuity requires slower operation (time delay) or longer delays (for a given abnormal current) for the protective devices that are closer to the power source. Opening only the protective device nearest (upstream) of the fault or overload and leaving the rest of the system operational is referred to as “selective coordination” (or just “coordination”), between protective devices.

The above discussion shows that maximum protection and maximum service continuity are somewhat inconsistent goals. The power system design engineer will often have to make suitable compromises between the two goals.

General Procedures

The general procedures for performing a coordination study include the following: (1) preparing an accurate one-line diagram, (2) determining the equipment protection guidelines, (3) selecting a plotting scale, (4) plotting the fixed points, (5) plotting/tracing the protective devices, and (6) analyzing the coordination study results.

One-Line Diagrams

Similar to many other types of power system studies, the coordination study procedures begin by preparing an accurate one-line diagram. As a minimum, the following data should be included on the one-line diagram: types, ratings, and settings of all protective devices; load, conductor, transformer, and motor data; and short circuit current values (symmetrical, asymmetrical, and X/R ratios). Figure 15 shows a one-line diagram that is being used to coordinate the time-overcurrent relays that are protecting a 3.75 MVA power transformer.

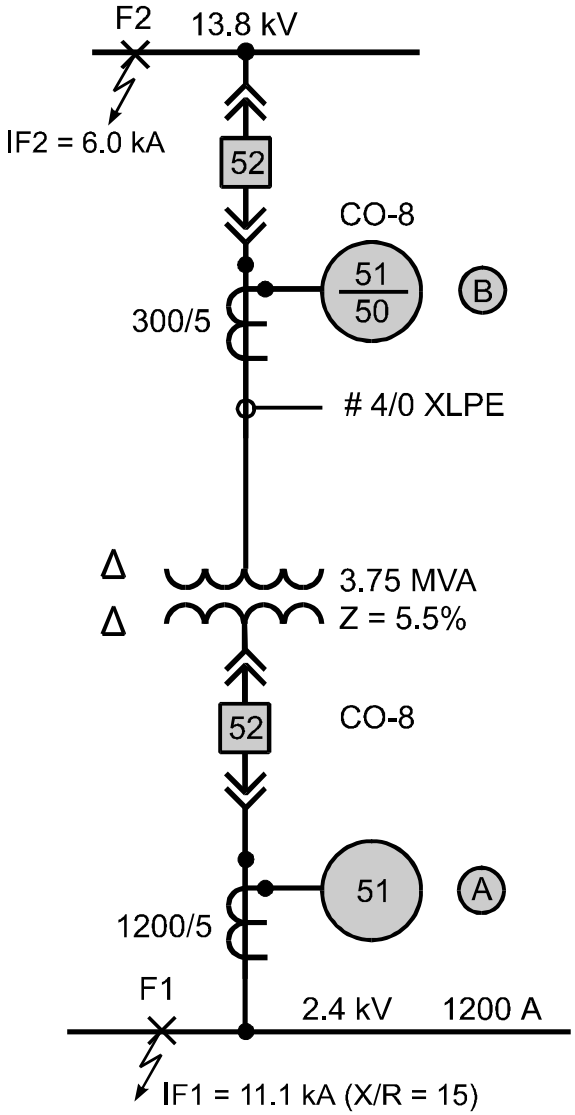


Figure 15. Example One-Line Diagram

Scale Selection Procedures

The response curves of all protective devices are plotted on common graphs so that they may be compared at all current and time points. The standard method that is used to plot device time/current characteristics (TCC) is to plot the devices on 4.5 x 5-cycle log-log graph paper (Figure 16).

The horizontal axis, which represents current, ranges from .5 to 10,000 amperes. The vertical axis, which represents time, ranges from .01 to 1000 seconds and/or .6 to 60,000 cycles. Because current limiting fuses and molded case circuit breakers may operate in less than 0.5 cycles (.00835 seconds), manufacturers of these devices may reproduce TCC curves with 7-cycle vertical scales, with times ranging from .001 to 10000 seconds (.06 to 600,000 cycles). The horizontal current scale also is often “shifted” for a particular plot by multiplying the standard current scale by a factor of 10, 100, or 1000 (x10, x100, x1000).

The coordination engineer should examine the range of currents that are to be plotted on the log-log graph paper. Generally, the ampere rating of the smallest device is the limiting factor on the left side of the paper, and the maximum available fault current is the limiting factor on the right side of the paper. The engineer should then select a scale that requires the least amount of calculation and manipulation. Usually, a current scale corresponding to the voltage level which has the most devices to be coordinated is selected as the scaling factor.

There will be as many scaling factors as there are voltage levels in the system. Once the initial scaling factor has been selected at a corresponding voltage level, the other scaling factor(s) must be calculated based on the voltage ratios (kV_p/kV_s or kV_s/kV_p) of the transformers. See Examples A and B.

Example A: A scaling factor of 100 @ 0.48 kV has been selected to plot the TCC curves for a particular coordination study. What are the scaling factors to plot the 4.16 and 13.8 kV TCC curves?

Answer A: **See Figure 17.**

Example B: A scaling factor of 10 @ 2.4 kV has been selected to plot the TCC curves for a particular coordination study. What are the scaling factors to plot the 0.208 and 13.8 kV TCC curves?

Answer B: **See Figure 18.**

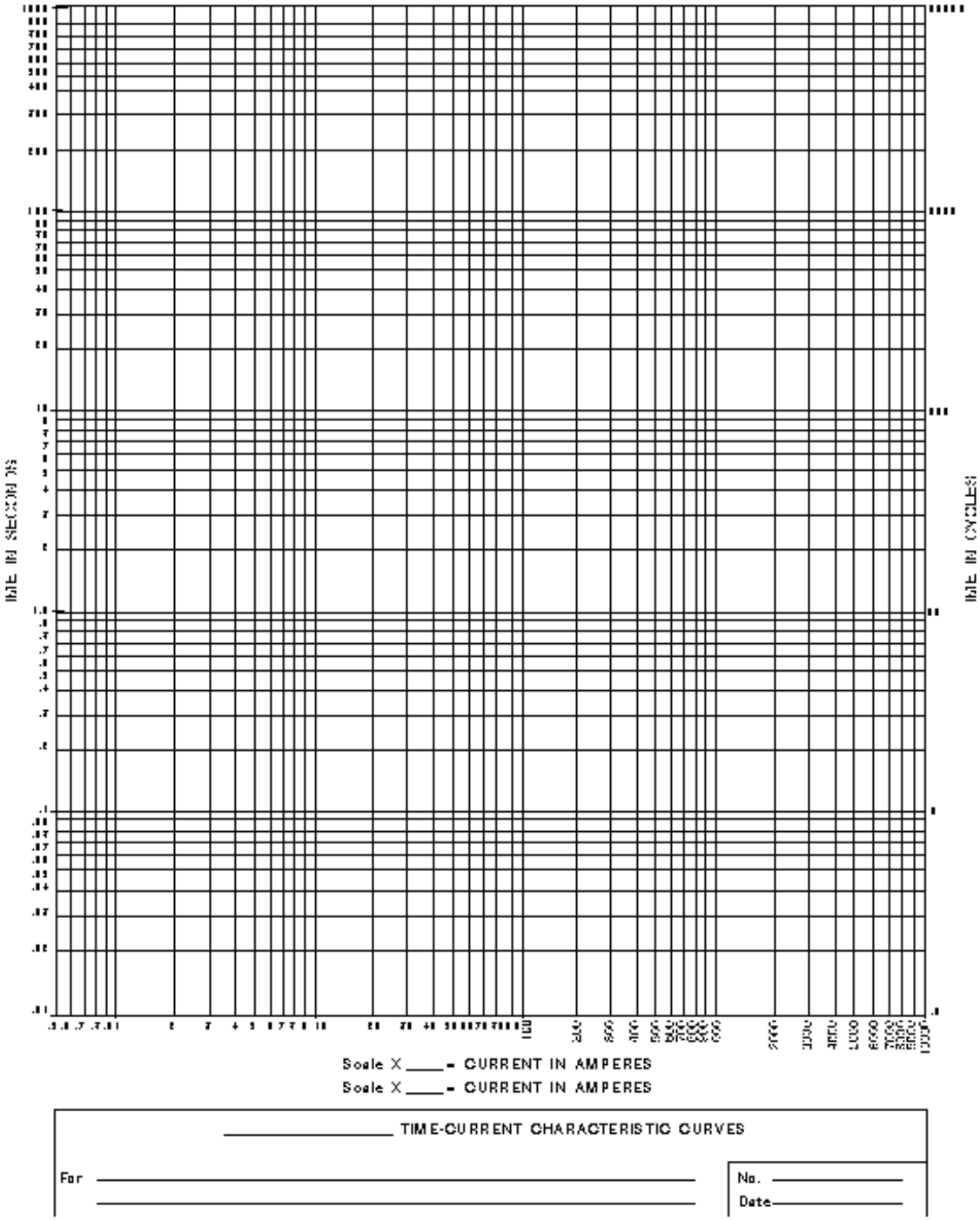


Figure 16. Typical Log-Log Paper

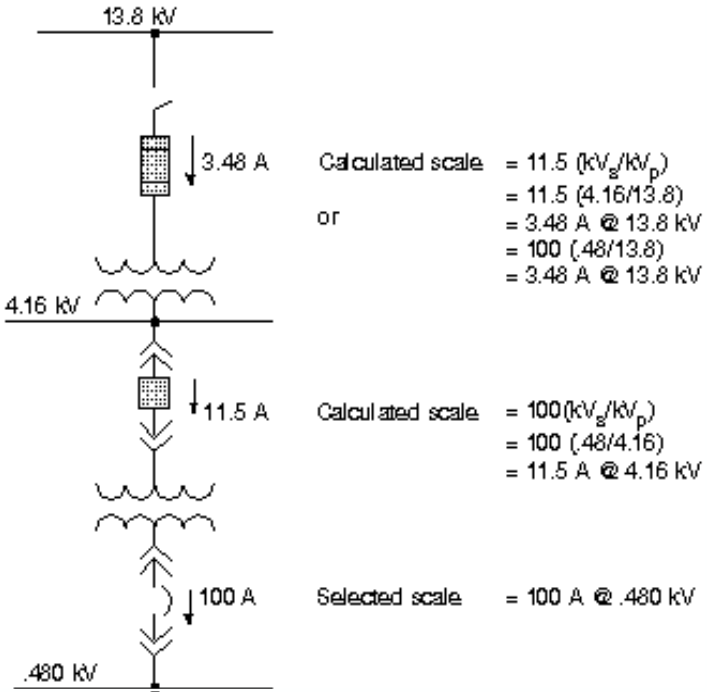


Figure 17. Example A Answer

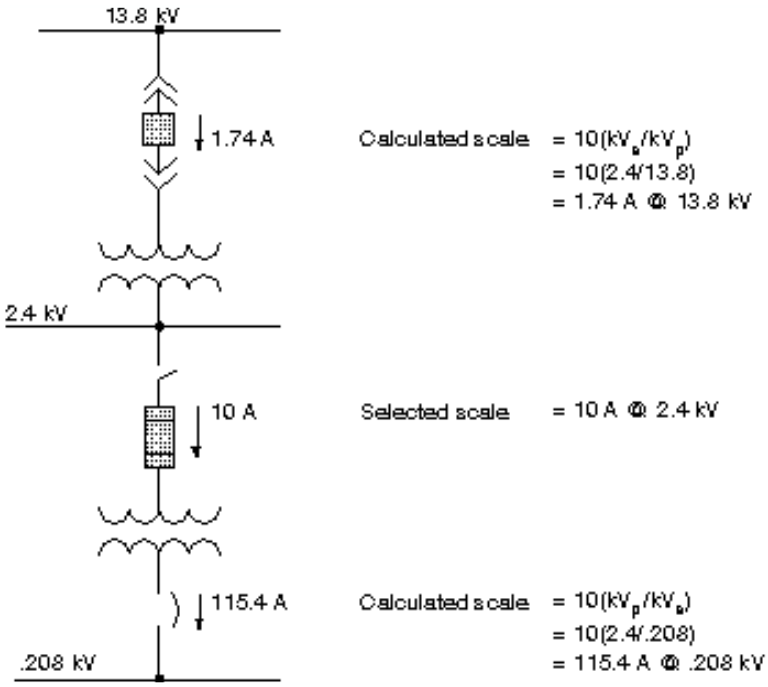


Figure 18. Example B Answer

Plotting of Fixed Points (Curves)

Fixed points (curves) are protection points and curves that do not change, regardless of the protective device ratings and settings. As a minimum, the following points and curves should be plotted on the log-log graph paper:

- Motor starting curves
- Motor thermal damage curves
- Transformer damage (Z) curves
- Transformer inrush point
- Cable damage curves
- Short circuit maximum fault points
- Cable ampacities
- NEC maximum protection points for motors, transformers, and cables

Protective Device Plotting/Tracing

In general, it is best to begin plotting the branch circuit's protective device TCC curves and to work toward the source. Stated another way, the TCC curves should be plotted left-to-right on the log-log coordination paper (plot "downstream" to "upstream"). When coordinating one device with many downstream devices, the upstream device should be set to coordinate with the largest or highest set downstream device. The upstream device will then automatically coordinate with all smaller downstream devices.

Selection of Ratings and Settings

The selection of ratings and settings of overcurrent protective devices to provide system protection and selective operation is often a trial and error process. For the best system protection, the smallest overcurrent device current rating that will allow normal load currents to flow, including any permissible overloads, should be selected. For selective coordination of the overcurrent protective devices, the TCC curves should be

adjusted as far to the left as possible without overlapping or crossing another curve.

Analysis of the Coordination Study

Coordination is not an exact science. Very often, a compromise between protection and coordination must be made, and some overlap of TCC curves may be necessary for purposes of protection. However, a careful analysis of the completed study will, as a minimum, let both the engineers and technicians know where coordination of the system has been compromised for the sake of equipment protection.

Data Requirements

Power Company Settings

Although the settings and ratings of the power company's protective devices are their responsibility, it is often helpful to know the rating, setting, and type of the first upstream power company protective device.

Transformer Data

As a minimum, the following transformer data are required to perform a coordination study:

- kVA ratings (OA/FA)
- Primary and secondary voltages
- Connections (e.g., wye-delta and delta-wye)
- Percent impedance (Z%)
- Liquid-filled or dry-type
- Overload capacities (capability)
- ANSI/IEEE damage curves (Z-curves)

Motor Data

As a minimum, the following motor data are required to perform a coordination study:

- Type (synchronous or induction)

- Horsepower (hp)
- Power factor (p.f.)
- Full load and locked-rotor amperes (FLA and LRA)
- Transient reactance (X%)
- Service factor (S.F. = 1.0 or 1.15)
  Saudi Aramco only specifies 1.0 S.F. motors.
- Starting (t_s) and locked-rotor (t_{LR}) (stall) times
- Starting type (e.g., full voltage, reduced voltage, etc.)
- Thermal damage (capability) curve

Load Data

As a minimum, the maximum load data, as well as any special load considerations, are required to perform a coordination study. For example, the expected normal and emergency loading conditions should be known to perform the study.

Fault Currents Available

The maximum symmetrical (I_{sym}) and asymmetrical (I_{asy}) fault currents, as well as the system X/R ratios at each protective device location, are required to perform a coordination study.

Conductor Data

As a minimum, the following conductor data are required to perform a coordination study:

- Material type (copper or aluminum)
- Conductor configuration (3-1/C or 1-3/C)
- Type insulation (e.g., 600 V, THWN or 15 kV, XLPE)
- Temperature ratings (e.g., 60 ^{o}C, 75 ^{o}C, and 90 ^{o}C)
- Type of conduit (e.g., steel and plastic)
- Ampacity ratings (NEC Article 310)
- Number per phase (e.g., 2/ϕ and 3/ϕ)
- Cable damage curves

Protective Device Data

Although there are many different protective devices that are used in an electrical power system, this Module will limit the discussion to the following protective devices:

- Molded Case Circuit Breakers (MCCBs)
- Low Voltage Power Circuit Breakers (LVPCBs)
- Medium Voltage Power Circuit Breakers (MVPCBs)
- Medium Voltage Fuses
- Overcurrent Relays

Molded Case Circuit Breakers (MCCBs) - The following listed MCCB data are required to perform a coordination study:

- Type and manufacturer (e.g., [W] Type HFB and GE Type NK)
- Frame size (e.g., 100 A and 225 A)
- Ampere trip ratings (e.g., 60 A and100 A)
- Adjustment ranges for large-frame size MCCBs (e.g., 5 to 10 or low-to-high)
- TCC curves (Figure 19)

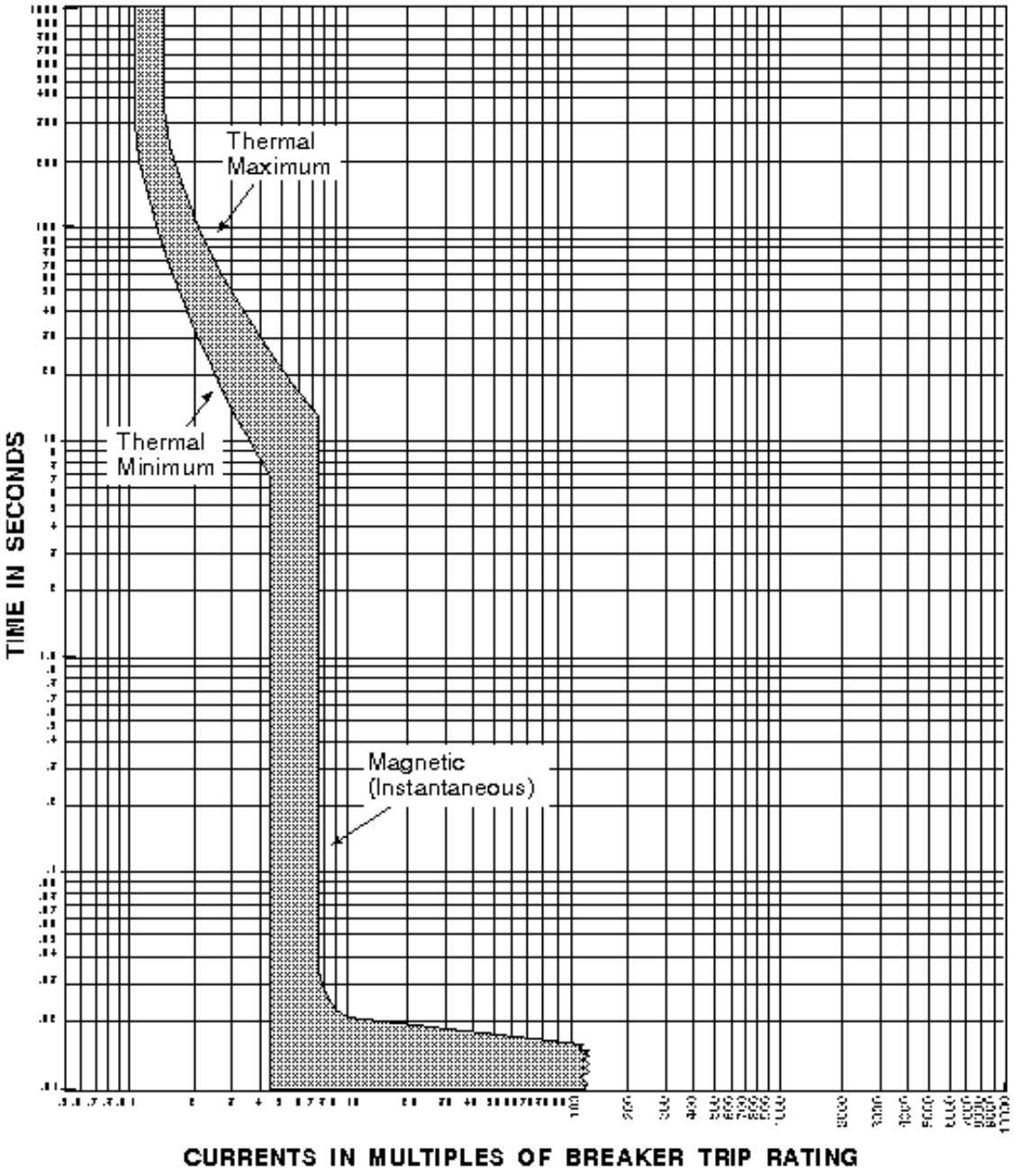


Figure 19. MCCB TCC Curve

Low Voltage Power Circuit Breakers (LVPCBs) - The following listed LVPCB data are required to perform a coordination study:

- Type and manufacturer (e.g., GE AK-25 and [W] DS-416)
- Frame sizes (e.g., 800 A and 1600 A)
- Trip unit type (e.g., GE Versatrip, (W) Amptector I-A, etc.)
- Ampere, sensor, and plug ratings (e.g., 800 A and 1200 A)
- Trip functions (e.g., long time, short time, and I^2t)
- Adjustment ranges (e.g., 0.5 - 1.0 I_n and 2 - 10x)
- TCC curves (Figure 20)

Medium Voltage Power Circuit Breakers (MVPCBs) do not have TCC curves. However, the type, manufacturer, and operating times (e.g., 3 cycles, 5 cycles, and 8 cycles) of the MVPCB are required to perform the coordination study.

Medium Voltage Fuse data requirements for use in a coordination study include the type (current limiting or non-current limiting), the manufacturer, the continuous current ratings, and the TCC curves (Figures 24 and 25).

Overcurrent Relay data requirements for use in a coordination study include the type (time-delay or instantaneous); the ampere tap (A.T.), time dial (T.D.), and instantaneous pickup (P.U.) adjustment ranges; the CT ratios; and the TCC curves (Figure 23).

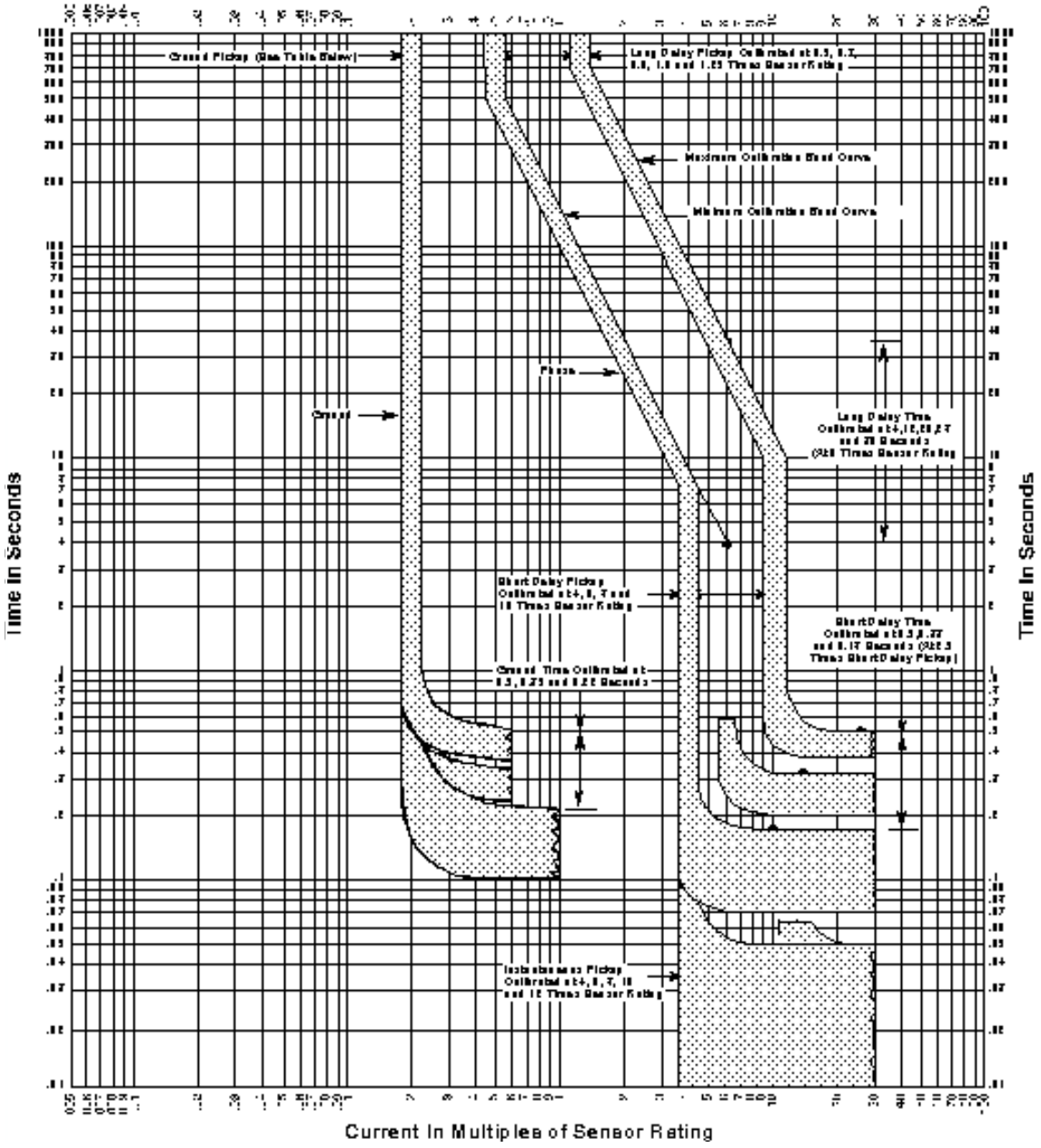


Figure 20. Amptector Trip Unit TCC Curves

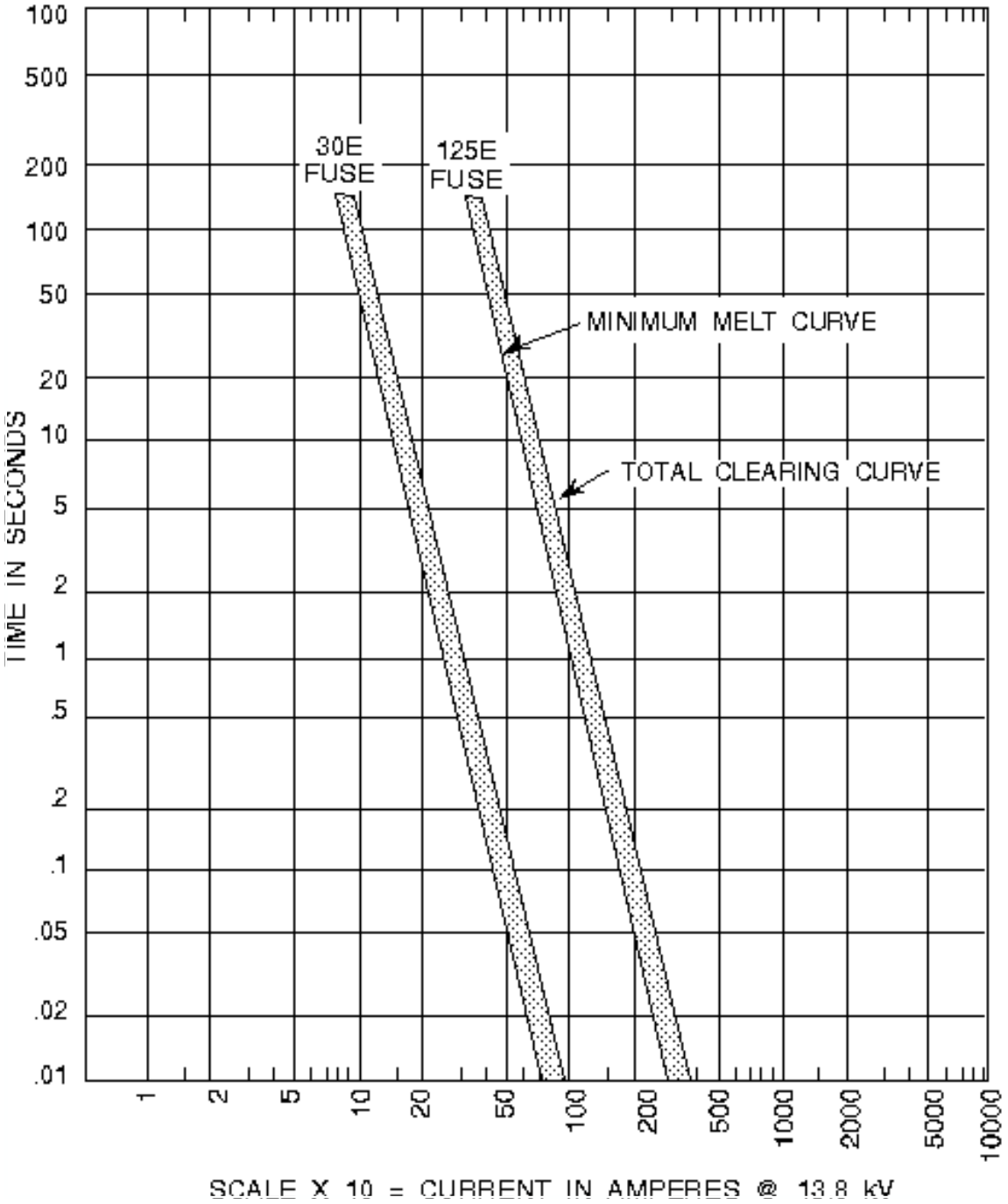


Figure 21. Medium Voltage Current Limiting Fuse TCC Curves

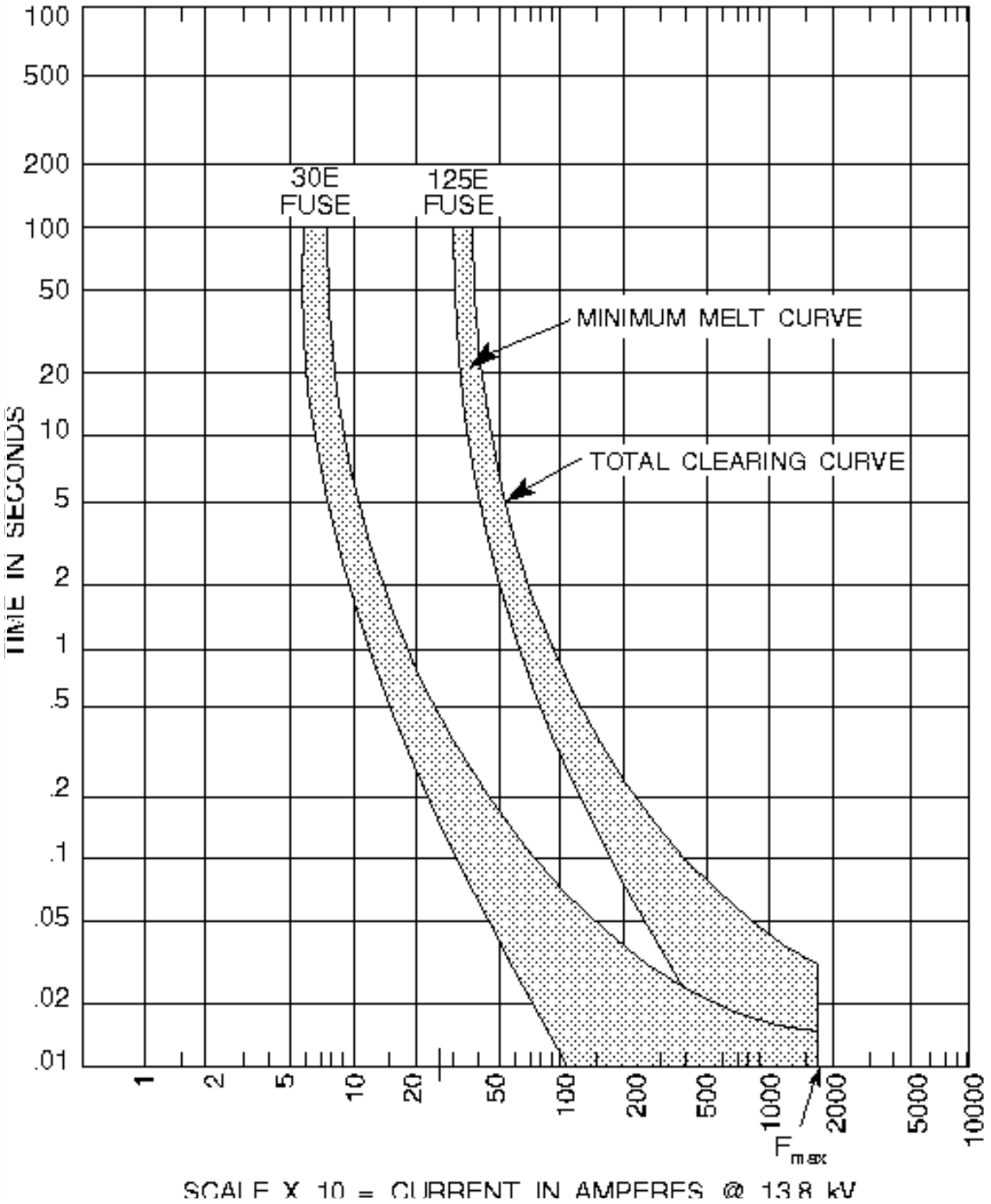


Figure 22. Medium Voltage Non-Current Limiting (Expulsion) Fuse TCC Curves

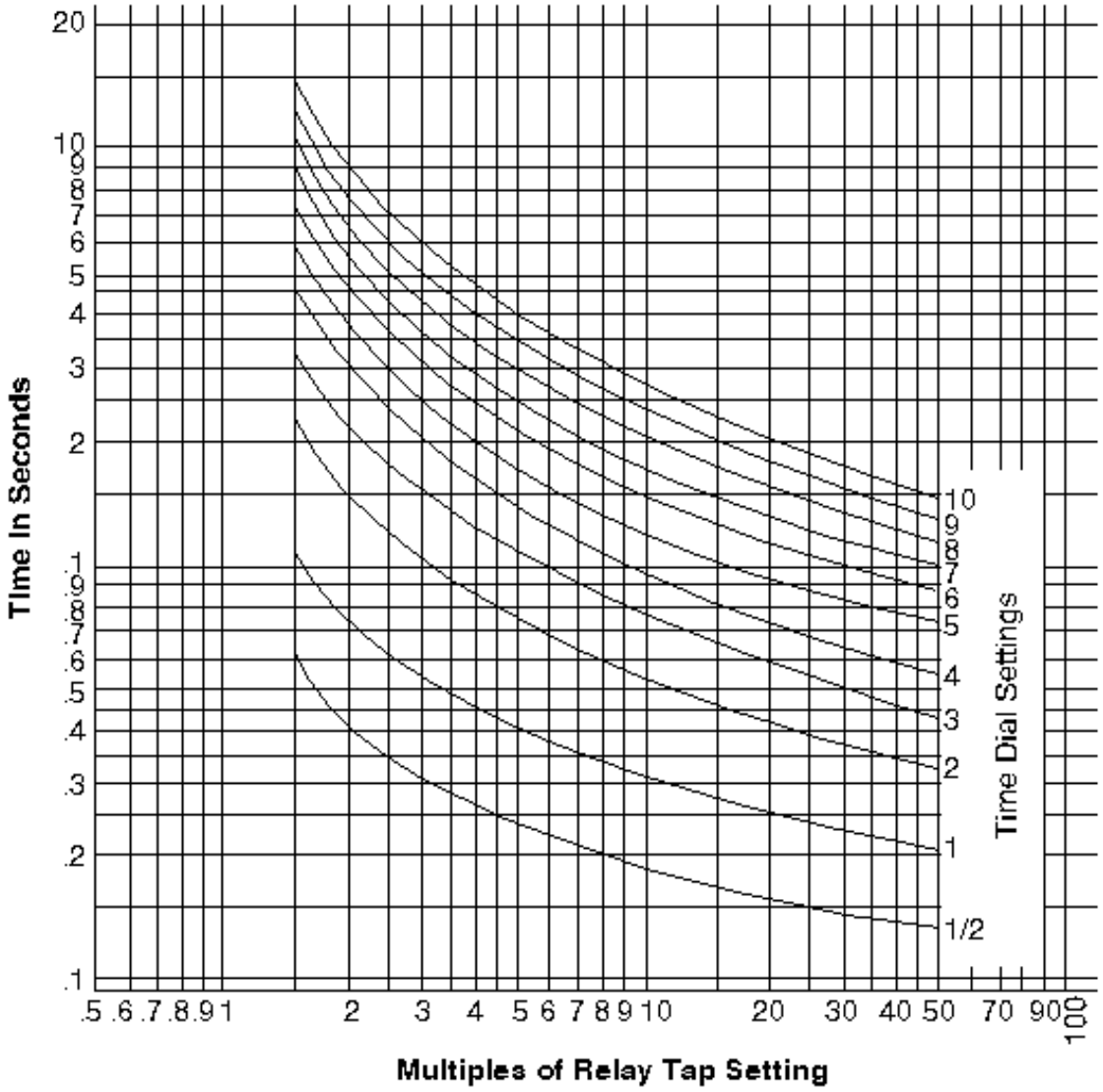


Figure 23. General Electric Type IAC51 Time Overcurrent Relay TCC Curves

GLOSSARY

A. T. (relay)	Ampere Tap
air-magnetic breaker	A type of medium voltage circuit breaker that has its contacts in air. A powerful electromagnet built into the arc chutes aids in extinguishing the arc.
analog (data)	Data in the form of continuously variable physical quantities such as voltages, currents, and resistances.
ANSI	American National Standards Institute
asymmetrical current (I_{asy})	A current where the envelopes of the peaks of the current waves are not symmetrical about the zero axis. Most short circuit currents are nearly always asymmetrical during the first few cycles after the fault occurs.
basic impulse level	A factory test that shows how well an insulation system can withstand a high voltage surge.
BIL	Basic Impulse Level
bus	A conductor or group of conductors that serves as a common connection for two or more circuits.
cable	A stranded conductor or a combination of conductors that is insulated from each other.
circuit breaker	A mechanical switching device that is capable of making, carrying, and breaking currents under normal and abnormal circuit conditions.
clearing time (t_c)	The amount of time that it takes a fuse to interrupt a circuit at a certain current level.
conductor	The current carrying element of a branch or feeder circuit. A conductor is usually a cable, an overhead line, or a bus duct.
conduit	A metallic or non-metallic tube that is used to mechanically protect (enclose) electric wires and cables. See raceway.
continuous current rating	The amount of current that a device can allow to pass through it without causing excessively high temperature or equipment failure.

current limiting fuse	A type of fuse that interrupts a fault current, but limits it to some value usually well below the peak current, and that operates in one-half cycle (.008 sec) or less.
current transformer (CT)	An instrument transformer that has its primary winding connected in series with the conductor carrying the current that is to be measured or controlled.
differential relay	A relay that by its design or application is intended to respond to the difference between incoming and outgoing electrical quantities associated with the protected equipment.
forced-cooled rating (FA)	A kVA rating that is specified on an oil-filled transformer. This FA rating is the transformer capacity with fans operating.
frame size	A term that describes the maximum continuous current rating, in amperes, of a circuit breaker.
full load amperage (I_{FLA})	The current that is drawn by a motor under full load conditions: for example, rated horsepower and rated voltage.
fuse	An electrical device that is designed to interrupt a circuit on an overload or a fault.
horsepower (hp)	The mechanical output power rating of a motor. One (1) hp equals 746 watts.
IEEE	Institute of Electrical and Electronics Engineers
induction disc relay	A form of relay armature in the shape of a disc that usually serves the combined function of providing an operating torque, by its location within the fields of an electromagnet that is excited by the input quantities, and a restraining force by motion within the field of a permanent magnet.
induction motor	A motor in which the field is produced by induction from the stator rather than from a direct current (dc) field winding.
instantaneous relay	A qualifying term that is applied to a relay or other device indicating that no delay is purposely introduced in its action.
instrument transformer	A transformer that is intended to reproduce in its secondary circuit, in a definite and known proportion, the current or voltage of its primary circuit, with its phase relations substantially preserved.

instruments	A term that describes a device that is used to measure or display a quantity under observation. Examples of instruments include a voltmeter or an ammeter.
inverse time relay	A relay in which the input quantity and operating time are inversely related throughout at least a substantial portion of the performance range. Types of inverse-time relays are frequently identified by such modifying adjectives as definite minimum time, moderately, very, and extremely to identify relative degree of inverseness of the operating characteristics of a given manufacturer's line of such relays.
inverse time-current (curve)	A term that is used to describe a TCC curve for a fuse, breaker, or protective relay. The curve indicates that as the current increases the time decreases.
locked-rotor amperage (I_{LRA})	The current that is drawn by a motor during starting. Also called starting current.
lockout relay	An electrically reset or hand-reset relay that holds associated devices inoperative until the relay is reset.
low voltage (LV)	Voltage levels that are less than 1000 volts. Usually called utilization level voltages.
low voltage switchgear	Switchgear rating that is less than or equal to 600 volts.
medium voltage switchgear	Switchgear rating that is between 601 and 15,000 volts.
medium voltage (MV)	Voltage levels that are greater than or equal to 1000 volts and that are less than 100,000 volts; usually called distribution level voltages.
melting time (t_m)	The amount of time that it takes a fuse element to melt at a certain level of current.
National Electric Code (NEC)	An electrical safety code developed and approved every three years by the National Fire Protection Association (NFPA 70).
nominal system voltage	A nominal assigned value that designates a system of a given voltage class.
non-current limiting fuse	A type of fuse that does not limit the peak fault current.

normally closed (N.C.) breaker	A breaker in which the current-carrying components are in engagement (closed) when the operating unit is in its normal position.
normally open (N.O.) breaker	A breaker in which the current-carrying components are not in engagement (not closed) when the operating unit is in its normal position.
opening time	The amount of time that it takes for a medium voltage breaker to open. Opening time is usually measured in cycles. One cycle equals 1/60 of a second (0.0167 seconds).
overcurrent protective device	An electrical device that is inserted in a circuit to protect the circuit against damage from an overload or short-circuit. The protection is achieved by automatic interruption.
P. U. (relay)	Pickup
polarity (marks)	The identification marks that are used to indicate the relative instantaneous polarities of the primary and secondary currents and voltages.
potential transformer (PT)	See voltage transformer.
power factor (p.f.)	The term cosine theta, where theta (θ) is the angle between the voltage and current waveshapes.
primary selective	A type of substation bus configuration that feeds two transformers from the same bus.
protective relay	A special relay that is designed to sense abnormal conditions in an electrical system.
raceway	Any channel that holds wires, cables, or bus bars. It may be metallic or non-metallic. Examples are conduit, cable duct, and cable tray. See conduit.
relay (general)	An electric device that is designed to interpret input conditions in a prescribed manner, and after specified conditions are met, to respond to cause contact operation or similar abrupt changes in associated electric control circuits. Relay inputs are usually electric but may be mechanical, thermal, or other quantities.
reliability	The ability of a substation or a piece of electrical equipment to operate without failure.

RTD	Resistance Temperature Detector.
secondary selective	A type of substation bus configuration that allows flexibility on a secondary bus system to feed all loads from one transformer by closing of a normally open (N.O.) tiebreaker.
self-cooled rating (OA)	A kVA rating that is specified on an oil-filled transformer. The kVA capacity (rating) of a transformer without the use of any additional cooling methods, such as fans.
short circuit current (I_{SC})	The current (usually very large) that flows in an electrical system as the result of a three-phase, phase-to-phase, double-phase-to-ground, or single phase-to-ground fault.
short-time rating	A rating for low voltage power circuit breakers (LVPCBs) and medium voltage power circuit breakers (MVPCBs) that describes the breaker's ability to withstand a fault current for a period of time. If a breaker does not have an instantaneous trip unit, it must have a short-time rating. The short-time rating of an LVPCB is 30 cycles (0.5 seconds) and the short-time rating of an MVPCB is 3 seconds.
substation	A group of electrical equipment items that has a power transformer rated 501 kVA or larger.
subtransient reactance ($X"_d$)	The apparent reactance of the stator winding at the instant a short circuit occurs. $X"_d$ determines the short-circuit current flow during the first few cycles after a fault occurs ($t < 3\sim$).
switchgear	A general term that describes switching and interrupting devices and their combination with associated control, instrumentation, metering, protective and regulating devices, assemblies of these devices with associated interconnections, accessories and supporting structures that are used primarily in connection with the generation, transmission, distribution, and conversion of electric power.
symmetrical current (I_{sym})	A current where envelopes of the peak of the current waves are symmetrical about the zero axis.
synchronous motor	A motor that has a field excited by direct current (dc) and a stator winding and in which alternating current (ac) flows. A type of ac motor that operates at synchronous speed.
T.B (relay)	Tap Block

T. D. (relay)	Time Dial
TCC	Time/Current Characteristic
time/current curves	Curves that show the operating characteristics of a protective device. The vertical axis shows time and the horizontal axis shows current. The curves are usually plotted on semi-log (4.5 x 5-cycle) paper.
voltage transformer (VT)	An instrument transformer that is intended to have its primary winding connected in shunt with a power supply circuit, the voltage of which is to be measured or controlled. Formerly called potential transformer (PT) in the USA.